

72nd BPSC PRELIMS

प्रहार

Test Series

Test-23/7

FULL LENGTH TEST

Solution

English



BPSC TEST-23/7 (SOLUTION)

Q 1. Ans. C

Explanation:

- The SI unit of Electric Resistance is Ohm (Ω). The SI unit of Electrical Resistivity (Specific Resistance) is Ohm-meter.
- **Capacitance** is measured in Farad (F).
- **Magnetic Flux** is measured in Weber (Wb).
- **Electric Charge** is measured in Coulomb (C).

Additional Fact:

- **Resistance vs. Resistivity:** Resistance depends on the length and cross-sectional area of a conductor, whereas Resistivity is an intrinsic property of the material and its temperature (independent of shape or size).
- **Magnetic Flux vs. Flux Density:** While Magnetic Flux is Weber, Magnetic Flux Density is measured in Tesla (T) or Wb/m².
- **CGS Conversions (Frequently asked):** The CGS unit for magnetic flux is the Maxwell. 1 Weber = 10⁸ Maxwell. The CGS unit for magnetic flux density is Gauss. 1 Tesla = 10⁴ Gauss.
- **Coulomb / Ampere Link:** Electric Current is the rate of flow of charge. Therefore, 1 Ampere = 1 Coulomb/second. One Coulomb contains approximately 6.24×10^{18} electrons.
- **Farad Breakdown:** 1 Farad = 1 Coulomb / 1 Volt. Because a Farad is a very large unit, practical capacitance is usually measured in microfarads (uF) or picofarads (pF).

Q 2. Ans. C

Explanation:

- When a ball is caught, its momentum has to be brought to zero. The change in momentum remains the same whether the hands are kept rigid or moved backwards. By pulling the hands back while catching the ball, the **time of contact** between the ball and the hands increases. **According to the impulse-momentum theorem, Force X Time = Change in Momentum.** Since the change in momentum is constant, an increase in the time of contact results in a decrease in the force exerted on the hands. Therefore, pulling the hands backwards while catching a ball reduces the **force experienced**, making the catch safer and less painful. Hence, the correct answer is **Force experienced**.
- **Additional Fact:**

- ✓ **Impulse:** Impulse is a large force acting for a very short duration. It equals the total change in momentum. Its SI unit is Newton-second (N-s).
- ✓ **Newton's Second Law:** The fundamental definition is that Force equals the rate of change of momentum. The famous equation $F = ma$ is derived from this.
- ✓ **Other Impulse Applications:** High jumpers landing on a sand bed or cushioned mattress (increases time, reduces force). Vehicles are provided with shock absorbers. Glass and chinaware are wrapped in straw or paper before packing.
- ✓ **Momentum vs. Kinetic Energy:** Momentum is mass times velocity. If kinetic energy is doubled, momentum increases by a factor of root 2.
- ✓ **Conservation of Momentum:** If no external force acts on a system, total momentum remains constant. Recoil of a gun and rocket propulsion are based on the Law of Conservation of Momentum and Newton's Third Law.

Q 3. Ans. C

Explanation:

- The **mass** of a body remains constant irrespective of its location because mass is the amount of matter contained in the body. However, the **weight** of a body depends on the gravitational force exerted by the Earth, which decreases with increasing distance from the Earth's centre. Therefore, when a body is taken to a height above the Earth's surface, the value of acceleration due to gravity (**g**) decreases, causing its **weight to decrease**. Hence, the correct answer is **Weight decreases**.
- **Additional Facts:**
 - ✓ **Mass versus Weight:** Mass is a scalar quantity measured in kilograms and never changes. Weight is a vector quantity, essentially a force measured in Newtons. Weight equals mass multiplied by gravity.
- **Variation of gravity with Depth:** Gravity also decreases as you go deeper into the Earth. At the exact centre of the Earth, gravity is zero, making the weight of any body zero.
- **Variation of gravity at the poles versus the equator:** The Earth is not a perfect sphere. Gravity is maximum at the poles and minimum at the equator. If you move from the equator to the poles, your weight increases.

- **Earth's Rotation Effect:** The rotation of Earth decreases the value of gravity. This reduction is maximum at the equator and zero at the poles. If the Earth suddenly stopped rotating, the weight of objects at the equator would increase.
- **Elevator Physics:** If a lift moves upward with uniform acceleration, apparent weight increases. If it moves downward with uniform acceleration, the apparent weight decreases. If the lift cable breaks, resulting in a free fall, the apparent weight becomes zero.

Q 4. Ans. C

Explanation:

- **Statement 1 is correct:** This is one of the fundamental laws of reflection of light. It states that the angle made by the incident ray with the normal at the point of incidence is equal to the angle made by the reflected ray with the normal. This law holds true for all reflecting surfaces.
- **Statement 2 is incorrect:** A plane mirror forms a **virtual image**, not a real image. The reflected rays only appear to meet behind the mirror and do not actually converge. Therefore, the image cannot be obtained on a screen, which is a characteristic of a virtual image.
- **Statement 3 is correct:** Lateral inversion means that the left side of an object appears as the right side in the image and vice versa. This is a characteristic feature of images formed by plane mirrors.
- **Additional Facts:**
 - ✓ **Focal Length and Power:** The focal length of a plane mirror is infinity, and its optical power is zero.
 - ✓ **Magnification:** The linear magnification of a plane mirror is always exactly plus one. This means the image is exactly the same size as the object and is erect.
 - ✓ **Minimum Mirror Size:** To see their complete full-length image, a person requires a plane mirror that is at least half of their own height.
 - ✓ **Relative Speed:** If an object moves towards a plane mirror with a speed V , the image appears to approach the object with a relative speed of $2V$.
 - ✓ **Multiple Images:** If two plane mirrors are placed at an angle θ to each other, the number of images formed is generally 360 divided by θ minus one. For parallel mirrors facing each other, infinite images are formed.

Q 5. Ans. B

Explanation:

- In a **series circuit**, all the components are connected one after another, forming a single path for the flow of electric charge. Since there is only one path available, the **same current flows through every component** of the circuit. However, the voltage is divided among the components depending on their resistances, and the power dissipated by each component may also differ. Therefore, the quantity that remains the same through all components in a series circuit is the **current**.
- **Additional facts:**
 - ✓ **Series Circuit Characteristics:** Total resistance is the sum of individual resistances. If one component fails, the entire circuit stops working. Fuses and circuit breakers are always connected in series with the live wire for safety.
 - ✓ **Parallel Circuit Characteristics:** In a parallel circuit, the voltage remains the same across all components, but the current divides into multiple branches. The total equivalent resistance is always less than the smallest individual resistance in the circuit.
 - ✓ **Domestic Wiring:** Houses are wired in parallel circuits. This ensures all appliances receive the same standard voltage and allows them to operate independently without breaking the circuit if one is turned off.
 - ✓ **Measuring Instruments:** An ammeter measures current and is always connected in series because it has very low resistance. A voltmeter measures voltage and is always connected in parallel because it has a very high resistance.
 - ✓ **Power Dissipation:** In a series circuit, a component with higher resistance dissipates more power and glows brighter if it is a bulb. In a parallel circuit, a component with lower resistance draws more current and dissipates more power.

Q 6. Ans. B

Explanation:

- Weight is the gravitational force acting on a body due to the Earth's attraction. The weight (W) of an object is given by the product of its mass (m) and the acceleration due to gravity (g). Thus, **Weight = Mass $\times$ g** . The SI unit of weight is **Newton (N)** because weight is a force. While the **mass** of a body remains constant, its **weight** can change from place to place as the value of g varies.

● **Additional facts:**

- ✓ **Weight on the Moon:** The acceleration due to gravity on the Moon is one-sixth that of Earth. Therefore, an object's weight on the Moon is one-sixth of its weight on Earth, but its mass remains completely unchanged.
- ✓ **Zero Weight Conditions:** A body experiences zero weight, or weightlessness, in a freely falling elevator, inside an orbiting space station or satellite, and at the exact centre of the Earth, where the value of gravity is zero.
- ✓ **Measuring Instruments:** Mass is measured using a physical balance or beam balance, which compares an unknown mass with known standard masses. Weight is measured using a spring balance, which measures the actual downward gravitational pull.

- ✓ **Action and Reaction Accelerations:** Even though action and reaction forces are equal in magnitude, they do not produce equal accelerations unless the two bodies have the exact same mass.
- ✓ **Non-uniform circular motion:** In uniform circular motion, only centripetal acceleration acts towards the centre. In non-uniform circular motion, there is also a tangential acceleration, meaning the net force is not directed exactly towards the centre.
- ✓ **Pseudo Forces:** A pseudo force is an apparent force that acts on all masses whose motion is described using a non-inertial or accelerating frame of reference. Centrifugal force is a classic example of a pseudo force; it is not a true reaction to centripetal force.

Q 7. Ans. A

Explanation:

- **Statement 1 is correct:** According to Newton's **First Law of Motion**, an object continues to move with constant velocity unless acted upon by an unbalanced external force. Since velocity remains constant, acceleration is zero, and therefore, the net force acting on the object is also zero.
- **Statement 2 is correct:** A body can experience multiple forces that balance each other. In such a case, individual forces may be non-zero, but their resultant (net force) is zero. Since acceleration depends on the net force, the body can have zero acceleration despite the presence of non-zero forces. For example, a book resting on a table experiences gravitational force downward and an equal normal reaction upward.
- **Statement 3 is incorrect:** According to Newton's **Third Law of Motion**, action and reaction forces are equal in magnitude and opposite in direction, but they act on **different bodies**. For example, when a person pushes a wall, the wall exerts an equal and opposite force on the person.
- **Additional Facts:**
 - ✓ **First Law and Inertia:** Newton's first law gives the qualitative definition of force and introduces the concept of inertia. Inertia is an inherent property of matter and is directly proportional to the mass of the body.
 - ✓ **Second Law as the Real Law:** Newton's second law is considered the fundamental or real law of motion because both the first and third laws can be derived from it. It gives the quantitative measure of force.

Q 8. Ans. A

Explanation:

- **Statement 1 is correct:** According to the **law of conservation of mass**, mass can neither be created nor destroyed in a chemical reaction. A balanced chemical equation has an equal number of atoms of each element on both sides of the equation, thereby ensuring that the total mass of reactants equals the total mass of products.
- **Statement 2 is incorrect:** Oxidation and reduction always occur together in a **redox reaction**. When one substance loses electrons (oxidation), another substance gains those electrons (reduction). Therefore, oxidation and reduction are complementary processes and occur simultaneously.
- **Additional facts:**
 - ✓ **Law of Conservation of Mass:** This fundamental law of chemistry was established by Antoine Lavoisier in 1789. He is often called the father of modern chemistry.
 - ✓ **Oxidation Definition:** It is defined as the addition of oxygen, the removal of hydrogen, or the loss of electrons from a substance.
 - ✓ **Reduction Definition:** It is defined as the addition of hydrogen, the removal of oxygen, or the gain of electrons by a substance.
 - ✓ **Oxidising and Reducing Agents:** The substance that gets oxidised causes reduction in the other, so it acts as the reducing agent. Conversely, the substance that gets reduced acts as the oxidising agent.
 - ✓ **Examples:** Corrosion, such as the rusting of iron, and the rancidity of food containing fats

and oils, are everyday examples of oxidation reactions. The chemical formula for rust is hydrated iron three oxide.

Q 9. Ans. D

Explanation:

- **Statement 1 is correct:** The crystalline lens of the human eye is a **convex lens**. It helps focus light rays onto the retina. The curvature of this lens can be adjusted by the ciliary muscles to focus on objects at different distances, a process known as accommodation.
- **Statement 2 is correct:** The eye lens forms a **real, inverted, and diminished image** of the object on the retina. The retina acts as a screen on which the image is formed. The brain then interprets the signals received from the retina, allowing us to perceive the object in its correct orientation.
- **Statement 3 is correct:** Myopia (short-sightedness) is a defect of vision in which a person can see nearby objects clearly but distant objects appear blurred. myopia is corrected by using a **concave (diverging) lens**, which diverges the incoming light rays so that the image of a distant object is formed on the retina.
- **Additional facts:**
 - ✓ **Hypermetropia:** Also known as farsightedness. Distant objects are clear, but nearby objects are blurred because the image forms behind the retina. It is corrected using a convex lens.
 - ✓ **Presbyopia:** With ageing, the ciliary muscles weaken, and the eye lens loses flexibility, making it hard to focus on nearby objects. It is corrected using bifocal lenses containing both concave and convex lenses.
 - ✓ **Astigmatism:** A defect where a person cannot simultaneously focus on horizontal and vertical lines due to an irregular curvature of the cornea. It is corrected using cylindrical lenses.
 - ✓ **Near Point and Far Point:** For a normal, healthy adult eye, the near point or least distance of distinct vision is 25 centimetres. The far point is infinity.
 - ✓ **Photoreceptors:** The retina contains two main types of light-sensitive cells. Rod cells respond to the intensity of light and are crucial for twilight or night vision. Cone cells respond to colour and function in bright daylight.

Q 10. Ans. C

Explanation:

- **Statement 1 is correct:** the pH scale measures the

concentration of hydrogen ions (H^+) in a solution. A decrease in pH value corresponds to an increase in hydrogen ion concentration, making the solution more acidic.

- **Statement 2 is correct:** bleaching powder ($CaOCl_2$) is used for disinfecting drinking water and killing harmful microorganisms. It is also used as a bleaching agent in the textile and paper industries.
- **Statement 3 is incorrect:** Washing soda is sodium carbonate decahydrate ($Na_2CO_3 \cdot 10H_2O$). Sodium hydrogen carbonate ($NaHCO_3$) is known as baking soda, not washing soda.
- **Additional Facts:**
 - ✓ **Plaster of Paris:** Chemically, it is calcium sulphate hemihydrate. When mixed with water, it changes into gypsum, setting into a hard solid mass. It is primarily used by doctors to support fractured bones in the right position.
 - ✓ **Biological pH Range:** The human body works within a narrow pH range of 7.0 to 7.8. When the pH of rainwater is less than 5.6, it is classified as acid rain, which can lower the pH of river water and threaten aquatic life.
 - ✓ **Tooth Decay Mechanism:** Tooth decay starts when the pH of the mouth falls below 5.5. Bacteria in the mouth produce acids by degrading sugar and food particles. Tooth enamel, made of calcium hydroxyapatite, is the hardest substance in the body but corrodes under these acidic conditions.
 - ✓ **Antacids:** During indigestion, the stomach produces excess hydrochloric acid, causing pain and irritation. Mild bases called antacids, such as Milk of Magnesia or magnesium hydroxide, are used to neutralise this excess acid.
 - ✓ **Baking Powder versus Baking Soda:** Baking powder is a mixture of baking soda and a mild edible acid, like tartaric acid. When heated during baking, it produces carbon dioxide gas, which gets trapped in the dough, causing bread or cake to rise and become soft and spongy.
 - ✓ **Natural Acids in Food:** Vinegar contains acetic acid, oranges contain citric acid, tamarind contains tartaric acid, tomatoes contain oxalic acid, and sour milk or curd contains lactic acid. Ant and nettle stings inject methanoic acid.

Q 11. Ans. C

Explanation:

- **Statement 1 is correct:** Carbon has the unique ability to form strong covalent bonds with other

carbon atoms. This property, known as **catenation**, enables carbon atoms to form long chains, branched chains, and rings, resulting in a vast number of carbon compounds.

- **Statement 2 is incorrect:** Carboxylic acids are acidic in nature and can release H^+ ions in aqueous solution. Alcohols, on the other hand, do not ionize appreciably to release H^+ ions and are generally neutral. Therefore, alcohols and carboxylic acids do not exhibit acidic behavior of similar strength.
- **Statement 3 is correct:** Unsaturated hydrocarbons contain double or triple bonds. These multiple bonds can break and allow other atoms or groups of atoms to add across them, making **addition reactions** a characteristic property of unsaturated hydrocarbons.
- **Additional Facts:**
 - ✓ **Allotropes of Carbon:** Diamond is the hardest natural substance where each carbon is bonded to four others in a rigid three dimensional structure. Graphite is soft and conducts electricity because each carbon is bonded to three others in hexagonal layers, leaving one free electron per carbon atom. Buckminsterfullerene contains 60 carbon atoms arranged in the shape of a football.
 - ✓ **Tetravalency:** Carbon has a valency of four, meaning it has four valence electrons in its outermost shell. This allows it to form single, double, or triple covalent bonds with other atoms.
 - ✓ **Saturated hydrocarbons:** Also known as alkanes. They contain only single bonds between carbon atoms. They are generally unreactive and undergo substitution reactions in the presence of sunlight, rather than addition reactions.
 - ✓ **Functional Groups:** The chemical properties of carbon compounds are determined by specific atoms or groups of atoms attached to the carbon chain. Common examples include the alcohol group, aldehyde group, ketone group, and carboxylic acid group.
 - ✓ **Esterification and Saponification:** A carboxylic acid reacts with an alcohol in the presence of an acid catalyst to form a sweet smelling substance called an ester. When an ester is treated with a strong base like sodium hydroxide, it converts back to an alcohol and a sodium salt of the carboxylic acid, which is the exact process used to manufacture soap.

Q 12. Ans. B

Explanation:

- Metals generally have **one, two, or three electrons in their outermost shell** and tend to **lose these electrons** to attain a stable electronic configuration. By losing electrons, a metal forms **positively charged ions (cations)**. For example, sodium loses one electron to form Na^+ while magnesium loses two electrons to form Mg^{2+} . This tendency to lose electrons is a characteristic property of metals and is responsible for the formation of ionic compounds.
- **Additional Facts:**
 - ✓ **Electropositivity:** Metals are known as electropositive elements precisely because they readily lose electrons. Nonmetals are electronegative because they tend to gain electrons to form negatively charged anions.
 - ✓ **Size of Cations:** When a neutral metal atom loses electrons to form a cation, the effective nuclear charge increases, and the outermost shell is often completely removed. Therefore, a cation is always strictly smaller than its neutral parent atom.
 - ✓ **Ionisation Energy:** The minimum amount of energy required to remove an electron from an isolated gaseous atom is called ionisation energy. Metals typically have very low ionisation energies, which explains why they lose electrons so easily.
 - ✓ **Ionic Bonding:** The strong electrostatic force of attraction between the positively charged metal cations and negatively charged non-metal anions results in the formation of ionic bonds, also known as electrovalent bonds. These compounds generally have high melting and boiling points.
 - ✓ **Aqueous Solutions:** When solid ionic compounds are dissolved in water, they dissociate into their constituent free ions. This mobility of ions is the reason why aqueous solutions of ionic compounds are excellent conductors of electricity.

Q 13. Ans. B

Explanation:

- In a **displacement reaction**, a **more reactive element** replaces a **less reactive element** from its compound. This occurs because the more reactive element has a greater tendency to lose or gain electrons and form compounds. For example, when zinc is added

to a copper sulfate solution, zinc displaces copper because zinc is more reactive than copper.

● **Additional Facts:**

- ✓ **Reactivity Series Order:** The decreasing order of reactivity for common metals is Potassium, Sodium, Calcium, Magnesium, Aluminum, Zinc, Iron, Lead, Hydrogen, Copper, Mercury, Silver, Gold.
- ✓ **Hydrogen Baseline:** Hydrogen is included in the metal reactivity series as a reference. Metals above hydrogen can displace it from dilute acids to release hydrogen gas. Metals below hydrogen, like copper and silver, cannot displace it.
- ✓ **Double Displacement:** In a double displacement reaction, two compounds react by mutually exchanging their ions to form two new compounds. This often results in the formation of an insoluble solid called a precipitate.
- ✓ **Thermite Reaction:** This is a highly exothermic displacement reaction where aluminium displaces iron from iron oxide. The reaction produces molten iron, which is widely used to weld cracked railway tracks.
- ✓ **Non-metal Displacement:** The displacement rule also applies to halogens. A more reactive non-metal displaces a less reactive one from its salt solution. For example, chlorine displaces bromine from an aqueous potassium bromide solution.

Q 14. Ans. C

Explanation:

- Across a period in the periodic table from **left to right**, the **atomic size generally decreases**. This is because, although electrons are added to the same outermost shell, the number of protons in the nucleus also increases. The increased nuclear charge attracts the electrons more strongly towards the nucleus, reducing the atomic radius. Thus, the size of atoms decreases progressively from left to right across a period.
- **Additional Facts:**
 - ✓ **Down a Group:** As we move down a group from top to bottom, the atomic size increases. This is because new electron shells are added, increasing the distance between the outermost electrons and the nucleus, which outweighs the effect of the increased nuclear charge.
 - ✓ **Metallic Character:** Metallic character, or the

tendency to lose electrons, decreases across a period from left to right and increases down a group. Conversely, non-metallic character increases across a period and decreases down a group.

- ✓ **Valency Trend:** Across a period from left to right, the valency of elements with respect to hydrogen first increases from one to four and then decreases to zero for noble gases. Down a group, all elements have the exact same number of valence electrons, so their valency remains constant.
- ✓ **Electronegativity:** The tendency of an atom to attract a shared pair of electrons is called electronegativity. It generally increases across a period, making fluorine the most electronegative element in the periodic table, and decreases down a group.
- ✓ **Ionisation Energy:** The energy required to remove an electron from an atom increases across a period due to the shrinking atomic size and stronger nuclear pull. It decreases down a group as electrons are further from the nucleus.
- ✓ **Noble Gas Exception:** The measured atomic radius actually jumps up slightly for noble gases at the extreme right of a period. Because noble gases do not form normal bonds, their size is measured using the van der Waals radius, which is inherently larger than the covalent radii used for other elements.

Q 15. Ans. A

Explanation:

- **Molarity (M)** is defined as the **number of moles of solute present in one litre of solution**. Molarity is **temperature-dependent** because the **volume of a solution changes with temperature** due to thermal expansion or contraction. As the temperature increases, the solution expands, increasing its volume and decreasing its molarity. Conversely, when the temperature decreases, the volume contracts and the molarity increases.
- **Additional Facts:**
 - ✓ **Molality:** Molality is defined as the number of moles of solute per kilogram of solvent. Because it depends only on mass and not on volume, molality is completely independent of temperature. For this reason, molality is often preferred over molarity for precise scientific measurements.

- ✓ **Normality:** Normality is defined as the number of gram equivalents of solute per litre of solution. Since it depends on volume, normality is also dependent on temperature.
- ✓ **Mole Fraction:** This is the ratio of the number of moles of one component to the total number of moles in the mixture. Like molality, mole fraction is based entirely on counts and masses, making it completely independent of temperature changes.
- ✓ **Concentration Terms:** Expressions like mass by mass percentage or parts per million by mass do not involve volume, so they are independent of temperature. Conversely, mass by volume percentage or volume by volume percentage are temperature dependent.
- ✓ **Dilution Principle:** When a solution is diluted by adding more solvent, the total number of moles of solute remains constant. The initial molarity multiplied by the initial volume always equals the final molarity multiplied by the final volume.

Q 16. Ans. A

Explanation:

- Oxidation is the process in which a substance **loses electrons**. According to the fundamental principle of electrochemistry, **oxidation always occurs at the anode**, while **reduction occurs at the cathode**. This is true for both galvanic (voltaic) cells and electrolytic cells, although the charge on the electrodes differs in these cells. Therefore, the correct answer is (A) **Anode**.
- **Additional Facts:**
 - ✓ **Charge of Electrodes:** In a galvanic cell or standard battery, the anode is the negative electrode because it generates electrons, and the cathode is positive. However, in an electrolytic cell, the anode is forced to be positive by an external power source, and the cathode is negative.
 - ✓ **Direction of Electron Flow:** In the external wire circuit, electrons always flow away from the anode and towards the cathode. Conventional electrical current flows in the exact opposite direction, from the cathode to the anode.
 - ✓ **Salt Bridge Function:** A salt bridge connects the two half cells in a galvanic cell. It completes the internal electrical circuit and maintains electrical neutrality by allowing the flow of non reacting ions, like potassium or nitrate,

without letting the two solutions physically mix.

- ✓ **Dry Cell Battery:** In a standard dry cell, also known as a Leclanche cell, the outer zinc container acts as the anode where oxidation occurs, and the central graphite rod acts as the cathode.

Q 17. Ans. B

Explanation:

- **The coordination number of a coordination compound is the number of donor atoms of the ligand that are directly bonded to the central metal ion or atom through coordinate (dative) bonds.** For example, in $[\text{Co}(\text{NH}_3)_6]^{3+}$, each NH_3 ligand donates one lone pair of electrons, so the coordination number is 6. In complexes containing bidentate or polydentate ligands, the coordination number is determined by the total number of donor atoms attached to the central metal ion, not simply the number of ligands.
- **Additional Facts:**
 - ✓ **Oxidation State:** This is distinct from the coordination number. The oxidation state is the theoretical charge the central metal ion would carry if all ligands were removed along with their shared electron pairs.
 - ✓ **Chelate Effect:** When a bidentate or polydentate ligand binds to a single metal ion using multiple donor atoms, it forms a ring-like structure called a chelate. Chelate complexes are generally much more stable than similar complexes formed with simple unidentate ligands.
 - ✓ **Denticity Examples:** Ammonia and water are unidentate ligands because they bind through a single atom. Oxalate and ethylenediamine are bidentate ligands. EDTA is a highly useful hexadentate ligand, meaning it binds a metal ion through two nitrogen and four oxygen atoms simultaneously.
 - ✓ **Coordination Geometry:** The spatial arrangement of the directly attached ligands defines the shape of the complex. A coordination number of 6 typically results in an octahedral geometry. A coordination number of 4 usually results in either a tetrahedral or a square planar geometry.
 - ✓ **Ambidentate Ligands:** These are unidentate ligands that have two different donor atoms and can bind through either of them. For

example, the nitrite ion can bind to a metal through either its nitrogen atom or its oxygen atom. Similarly, the thiocyanate ion can bind through its sulfur atom or its nitrogen atom.

Q 18. Ans. A

Explanation:

- **Statement 1 is correct:** asexual reproduction involves a single parent and does not involve the fusion of gametes. As a result, the offspring produced are genetically identical to the parent and to one another in most cases. Such offspring are known as clones, except for minor variations that may arise due to DNA copying errors.
- **Statement 2 is correct:** *Amoeba* reproduces asexually by **binary fission**, in which the nucleus divides first, followed by the division of the cytoplasm, producing two genetically identical daughter cells under favourable conditions.
- **Statement 3 is incorrect:** **spore formation is an asexual method of reproduction** observed in fungi such as *Rhizopus* (**bread mould**). The spores are produced inside sporangia and germinate into new individuals under favourable conditions. Therefore, spore formation is not a sexual method of reproduction.
- **Additional Facts:**
 - ✓ **Multiple Fission:** The malarial parasite *Plasmodium* reproduces by multiple fission, dividing into many daughter cells simultaneously inside a protective cyst.
 - ✓ **Budding:** *Hydra* and *Yeast* reproduce by budding. A small outgrowth or bud develops due to repeated cell division at one specific site, eventually detaching to form a new independent organism.
 - ✓ **Fragmentation:** *Spirogyra*, a filamentous green alga, simply breaks up into smaller pieces upon maturation. Each fragment grows into a new, complete individual.
 - ✓ **Regeneration:** Simple animals like *Planaria* and *Hydra* can regenerate into complete organisms if cut into multiple pieces. This is carried out by specialised cells, but is not considered true reproduction since most organisms cannot wait to be cut to reproduce.
 - ✓ **Vegetative Propagation:** Plants can reproduce asexually using vegetative parts like roots, stems, and leaves. Examples include *Bryophyllum* reproducing through leaf notches, potato through stem tubers, and sweet potato through

roots. This method is faster than growing from seeds and perfectly preserves the parent plant's genetic traits.

- ✓ **Definite Orientation Fission:** *Leishmania*, the pathogen that causes kala azar, has a whip-like structure at one end. Unlike *Amoeba*, which can divide in any plane, *Leishmania* undergoes binary fission in a definite longitudinal orientation.

Q 19. Ans. B

Explanation:

- **Pair 1 is correct:** insulin is secreted by the pancreas and helps lower the blood glucose level by promoting the uptake and storage of glucose in body cells.
- **Pair 2 is incorrect:** Adrenaline is secreted by the adrenal glands and prepares the body to respond to stressful situations by increasing heart rate, breathing rate, and blood glucose levels. It does not promote body growth.
- **Pair 3 is correct:** Thyroxine, secreted by the thyroid gland, regulates the metabolism of carbohydrates, proteins, and fats, thereby controlling the overall metabolic rate of the body.
- **Pair 4 is incorrect:** **Growth hormone, secreted by the pituitary gland**, primarily stimulates the growth of bones and body tissues. Raising blood glucose during stress is mainly the function of adrenaline, not growth hormone.

Q 20. Ans. D

Explanation:

- **Statement:** The nephron is the structural and functional unit of the kidney.
- **Conclusion I: Correct.** Each nephron filters blood, removes nitrogenous wastes, and helps in the formation of urine through the processes of filtration, selective reabsorption, and tubular secretion. Thus, it performs the basic excretory function of the kidney.
- **Conclusion II: Incorrect.** Bile is produced by the liver, not by the nephron. It is stored in the gall bladder and released into the small intestine, where it helps in the digestion and emulsification of fats. Therefore, bile production is not a function of the nephron.

Q 21. Ans. D

Explanation:

- **Statement:** Decomposers are an important part of ecosystems.
- **Conclusion I: Correct.** Decomposers such as bacteria

and fungi break down dead plants, animals, and organic wastes into simpler substances. In doing so, they recycle nutrients back into the environment, making them available for reuse by plants and maintaining nutrient cycles in ecosystems.

- **Conclusion II: Incorrect.** The conversion of solar energy into chemical energy is carried out by **green plants and other photosynthetic organisms** through the process of photosynthesis. Decomposers do not perform photosynthesis and therefore cannot convert solar energy directly into chemical energy.

Q 22. Ans. D

Explanation:

- In Mendel's experiments on pea plants, **dwarf stem** was identified as a **recessive trait**, while **tall stem** was the corresponding dominant trait. A recessive trait is expressed only when both alleles are recessive (homozygous). Similarly, **violet flower** is dominant over white flower, and **round seed** is dominant over wrinkled seed. Therefore, among the given options, **dwarf stem** is the recessive trait.

Q 23. Ans. C

Explanation:

- Double fertilization is a unique characteristic feature of **angiosperms (flowering plants)**. In this process, one male gamete fuses with the egg cell to form the **zygote (syngamy)**, while the other male gamete fuses with the two polar nuclei to form the **primary endosperm nucleus (triple fusion)**, which develops into the endosperm. Since two fertilization events occur in the same embryo sac, the process is called **double fertilization**.
- **Bryophytes:** These are the simplest land plants, such as mosses and liverworts. They reproduce by spores and require water for fertilization. **Double fertilization does not occur** in bryophytes.
- **Gymnosperms:** These are seed-producing plants with **naked seeds**, such as pine, deodar, and Cycas. Fertilization involves only one male gamete fusing with the egg cell. **Double fertilization is absent** in gymnosperms.
- **Pteridophytes:** These are vascular plants such as ferns and horsetails that reproduce by **spores** rather than seeds. They also require water for fertilization, and **double fertilization does not occur**.

Q 24. Ans. B

Explanation:

- **Statement:** Rhizobium lives in association with leguminous plants.

- **Conclusion I:** Incorrect. **Rhizobium is not a photosynthetic bacterium.** It does not manufacture food for the plant. Instead, it lives symbiotically in the root nodules of leguminous plants, where the plant provides it with food and shelter.

- **Conclusion II:** Correct. Rhizobium helps in **nitrogen fixation** by converting atmospheric nitrogen into usable nitrogen compounds such as ammonia. These compounds can be utilized by the plant for growth and development, thereby improving soil fertility.

Q 25. Ans. A

Explanation:

- Ribosomes are the **sites of protein synthesis** in the cell. They read the genetic information carried by messenger RNA (mRNA) and assemble amino acids in the correct sequence to form proteins.
- The **nucleus** stores genetic material and controls cellular activities, the **vacuole** is mainly involved in storage, and the **centriole** plays a role in cell division in animal cells.

Q 26. Ans. B

Explanation:

- GLP-1 receptor agonists are a class of medications that mimic the action of **glucagon-like peptide-1 (GLP-1)**, a natural hormone secreted by the intestines.
- In India, these medicines are **strictly regulated** and may be prescribed only by **endocrinologists, internal medicine specialists, and cardiologists**.
- They were originally developed and marketed by pharmaceutical companies such as **Novo Nordisk** and **Eli Lilly** for the treatment of chronic metabolic disorders.
- The primary indication of GLP-1 receptor agonists is the management of **Type 2 diabetes mellitus**, where they enhance **glucose-dependent insulin secretion** and improve blood glucose control.

Q 27. Ans. A

Explanation:

- **Statement:** The human heart has four chambers and supports double circulation.
- **Conclusion I: Incorrect.** The human heart has four chambers that keep **oxygenated and deoxygenated blood completely separate**. This separation ensures that oxygen-rich blood is efficiently supplied to the body and oxygen-poor blood is sent to the lungs for oxygenation. Therefore, the two types of blood do not mix in a healthy human heart.

- **Conclusion II: Incorrect.** Double circulation provides an efficient supply of oxygen to body tissues by ensuring complete separation of oxygenated and deoxygenated blood. This efficient oxygen transport supports the high energy requirements of humans. Therefore, double circulation does not maintain an inefficient oxygen supply.

Q 28. Ans. C

Explanation:

- NARIT-AI is a specialized, private artificial intelligence system developed to assist law enforcement officers in the investigation and prosecution of complex narcotics cases. Developed by the Gujarat Police (Western Railway Division) Unlike general-purpose AI, it operates within a secure, closed environment and is specifically trained on Indian legal statutes, judicial precedents, and the provisions of the NDPS Act. Its primary objective is to ensure that investigating officers strictly follow the procedural requirements of the Narcotic Drugs and Psychotropic Substances (NDPS) Act, where even minor procedural lapses can result in acquittals. NARIT-AI also provides real-time legal and analytical support, helping investigators build legally sound and evidence-based cases capable of withstanding scrutiny during court proceedings.

Q 29. Ans. B

Explanation:

- **Statement 1 is incorrect:** India's first indigenous Prototype Fast Breeder Reactor (PFBR) is located at Kalpakkam, Tamil Nadu, not Kerala. It is India's first indigenous PFBR and has recently achieved criticality, a major milestone in India's three-stage nuclear power programme. The 500 MWe PFBR has been developed by Bharatiya Nabhikiya Vidyut Nigam Ltd. (BHAVINI).
- **Statement 2 is correct:** A Fast Breeder Reactor (FBR) is an advanced nuclear reactor designed to generate more fissile fuel than it consumes while producing electricity. It uses fast (high-energy) neutrons to sustain the nuclear fission chain reaction, unlike conventional reactors, which rely on slow (thermal) neutrons. This breeding capability improves fuel utilization and supports long-term energy security.
- **Criticality** is the stage at which a self-sustaining nuclear chain reaction is established. At this point, the number of neutrons produced by fission exactly

equals the number of neutrons lost through leakage or absorption plus those that trigger subsequent fission reactions. Achieving criticality is a key milestone before a reactor begins power generation.

Q 30. Ans. C

Explanation:

- The Government of India has laid the foundation stone for the country's first 3D glass semiconductor chip packaging facility in Bhubaneswar, Odisha, marking a significant step toward strengthening India's advanced semiconductor ecosystem. 3D glass semiconductor packaging is an advanced form of 3D Heterogeneous Integration (3DHI) that uses glass substrates instead of conventional organic or silicon substrates to vertically stack and interconnect multiple chip components. This technology enables different types of chips, such as logic, memory, and sensors, to be integrated into a single compact, high-performance 3D module, improving efficiency, performance, and power consumption.

Q 31. Ans. D

Explanation:

- Nitish Kumar → First Indian CM to take oath ten times — World Book of Records
 - ✓ Nitish Kumar became the first Chief Minister in India to take oath ten times, earning recognition in the World Book of Records.
- Parveen Shaikh → Sanctuary Wildlife Service Award 2025
 - ✓ Parveen Shaikh, a noted wildlife rescuer and conservationist, was honoured with the Sanctuary Wildlife Service Award 2025 for her contributions to wildlife conservation.
- Samrat Rana → ISSF World Championship Gold, Cairo
 - ✓ Samrat Rana created by becoming the first Indian shooter to win an individual men's 10m Air Pistol World Championship gold medal at the ISSF World Championship in Cairo, Egypt.
- Rouble Nagi → Global Teacher Prize 2026
 - ✓ Rouble Nagi, a renowned artist and social worker, was awarded the Global Teacher Prize 2026 for her contributions to education and community development.

Q 32. Ans. B

Explanation:

- Gurugram, Haryana, recently launched India's first

QR code-based kiosk for digital payment of traffic challans. The initiative was introduced by the Gurugram Traffic Police to simplify and speed up the payment process for traffic violations.

- Under this system, offenders can scan a QR code at the kiosk and make instant payments through various digital modes such as UPI, net banking, and mobile wallets, reducing paperwork and improving transparency.

Q 33. Ans. B

Explanation:

- The IUCN World Conservation Congress 2025 was held in Abu Dhabi, United Arab Emirates (UAE). During the Congress, a motion was adopted recognising India's Dugong Conservation Reserve in Tamil Nadu as a significant conservation initiative for the protection of the endangered dugong and its seagrass habitat.
- The World Conservation Congress is the highest decision-making body of the International Union for Conservation of Nature and brings together governments, NGOs, scientists, indigenous groups, and conservation experts to shape global environmental policy.

Q 34. Ans. C

Explanation:

- **Statement 1 is correct:** The Belém Health Action Plan (BHAP) was launched at COP30 in Belém, Brazil, as the first international climate adaptation framework dedicated specifically to the health sector. The plan aims to strengthen the resilience of health systems against climate-related risks through improved surveillance, evidence-based policymaking, capacity building, and innovation.
- **Statement 2 is correct:** The Food Waste Breakthrough was officially launched at COP30 in Belém, Brazil, by the United Nations Environment Programme (UNEP) and its partners. The initiative seeks to halve global food waste by 2030, reduce methane emissions, and contribute to global food security and climate action.

Q 35. Ans. C

Explanation:

- According to the World Economic Forum's (WEF) Global Gender Gap Report 2025, Iceland ranked 1st in the Global Gender Gap Index 2025, retaining the top position for the 16th consecutive year. Iceland achieved a gender parity score of 92.6%, making it

the only country to have closed more than 90% of its gender gap.

- **The report evaluates 148 countries across four dimensions:**

- ✓ Economic Participation and Opportunity
- ✓ Educational Attainment
- ✓ Health and Survival
- ✓ Political Empowerment

- **Top 5 Countries in Gender Gap Index 2025**

- ✓ Iceland
- ✓ Finland
- ✓ Norway
- ✓ United Kingdom
- ✓ New Zealand
- ✓ India ranked 131st out of 148 countries in the Global Gender Gap Index 2025

Q 36. Ans. A

Explanation:

- **State of World Population 2025** → United Nations Population Fund (UNFPA)
 - ✓ The State of World Population is the flagship annual report of the UNFPA. It focuses on global demographic trends, reproductive health, gender equality, and population-related development challenges.
- **World Energy Investment Report 2025** → International Energy Agency (IEA)
 - ✓ The World Energy Investment Report is published annually by the IEA. It tracks global investment flows across the energy sector, including fossil fuels, electricity, renewables, grids, storage, and clean energy technologies.
- **Global Assessment Report on Disaster Risk Reduction 2025** → UN Office for Disaster Risk Reduction (UNDRR)
 - ✓ The Global Assessment Report (GAR) is the flagship report of the UNDRR. It provides a comprehensive assessment of global disaster risks and supports the implementation of the Sendai Framework for Disaster Risk Reduction.
- **Global Annual to Decadal Climate Update 2025–2029** → World Meteorological Organisation (WMO)
 - ✓ The Global Annual to Decadal Climate Update is released by the WMO. It presents near-term climate projections, including global temperature trends, rainfall patterns, and the likelihood of climate anomalies over the coming years.

Q 37. Ans. D

Explanation:

- The report titled “Chemical Industry: Powering India’s Participation in Global Value Chains” was released by NITI Aayog in July 2025. The report examines the current status of India’s chemical sector and recommends measures to enhance its integration into global value chains (GVCs).
- Key recommendations of the report include:
 - ✓ Establishing world-class chemical manufacturing hubs.
 - ✓ Improving port and logistics infrastructure.
 - ✓ Facilitating technology acquisition and innovation.
 - ✓ Streamlining environmental clearances.
 - ✓ Leveraging Free Trade Agreements (FTAs) to expand market access.

Q 38. Ans. A

Explanation:

- Gujarat has become the first state in India to launch a dedicated SpaceTech Policy, namely the SpaceTech Policy 2025–2030. The policy aims to position Gujarat as a leading hub for the space sector by promoting investment, innovation, research, manufacturing, and startups related to space technology.
- The policy seeks to attract private participation across the space value chain, including:
 - ✓ Satellite manufacturing
 - ✓ Launch vehicle components
 - ✓ Ground stations
 - ✓ Space applications and services
 - ✓ Research & Development (R&D)

Q 39. Ans. A

Explanation:

- **Statement 1 is correct:** Mumbai inaugurated the Mandale Metro Depot for Line 2B, marking it as Asia’s largest elevated metro depot. Spread across 30.45 hectares, the Mandale Depot has the capacity to house 72 eight-coach metro trains, with a double-level car shed designed to hold 36 trains on each level.
- **Statement 2 is incorrect:** The depot is located in Mankhurd, in Mumbai’s eastern suburbs (Maharashtra), not in Thane district. This 23.6-kilometre-long elevated Metro Line 2B connects DN Nagar (Andheri West) to Mandale (in Mankhurd East) of Mumbai. The depot was

developed by the Mumbai Metropolitan Region Development Authority (MMRDA).

Q 40. Ans. D

Explanation:

- West Bengal became the first state in India to secure a conviction in a digital arrest cyber fraud case. The Kalyani Court in Nadia district sentenced nine accused to life imprisonment for defrauding a retired agricultural scientist of nearly ₹1 crore through a fake digital arrest scheme. The verdict is considered a landmark development in India’s efforts to combat cybercrime and online financial fraud.

Q 41. Ans. B

Explanation:

- In August 2025, Sonali Mishra, a 1993-batch IPS officer from the Madhya Pradesh cadre, assumed charge as the Director General (DG) of the Railway Protection Force. This marked the first time a woman was appointed to lead the 143-year-old security force.
- She succeeds Manoj Yadava and will hold this position until her superannuation (retirement) on October 31, 2026.
- **RPF:** The Railway Protection Force was established in 1957 and was given the status of an Armed Force of the Union of India in 1985. It functions under the Ministry of Railways and is responsible for protecting railway property, passenger areas, and passengers.

Q 42. Ans. A

Explanation:

- India achieved the 20% ethanol blending (E20) target **five years** ahead of its original 2030 deadline. Ethanol blending in petrol has risen from just 1.5% in 2014 to 20% in 2025, a nearly 13-fold increase over 11 years. Ethanol production has surged from 38 crore litres in 2014 to 661.1 crore litres by June 2025, and as a result, India has saved approximately ₹1.36 lakh crore in foreign exchange by reducing its dependency on imported crude oil. Additionally, ₹1.96 lakh crore have been paid to distilleries and ₹1.18 lakh crore disbursed to farmers, while the increased use of ethanol-blended petrol has helped reduce carbon dioxide emissions by 698 lakh tonnes.

Q 43. Ans. C

Explanation:

- **Statement 1 is correct:** The Sagarmatha Sambaad is an international dialogue forum initiated by Nepal, and its inaugural edition was held in

Kathmandu, Nepal. The platform is named after Mount Sagarmatha (Mount Everest), symbolising the Himalayan region and its global environmental significance.

- **Statement 2 is correct:** The theme of the first Sagarmatha Sambaad was “Climate Change, Mountains, and the Future of Humanity.” It focused on the urgent issues of climate change impacts on mountain ecosystems, glacier melting, biodiversity loss, and sustainable development in mountainous regions, particularly the Himalayas.

Q 44. Ans. C

Explanation:

- Aastha Poonia recently made by becoming the first woman inducted into the Fighter Stream of Naval Aviation in the Indian Navy. This marks a significant milestone in the integration of women into combat aviation roles in India’s armed forces.
- The Fighter Stream involves operating combat aircraft from naval air stations and aircraft carriers, including high-intensity missions such as air defence, strike operations, and fleet protection.

Q 45. Ans. D

Explanation

- Friedrich Merz is a German politician and leader of the Christian Democratic Union (CDU). He was recently elected as the Chancellor of Germany, making him the head of government of the country.
- Germany is a federal parliamentary republic, where the Chancellor is the most powerful executive authority, elected by the Bundestag (Federal Parliament).

Q 46. Ans. A

Explanation:

- The World Food Prize 2025 was awarded to **Dr. Mariangela Hungria**, a Brazilian microbiologist known for her pioneering work in biological nitrogen fixation (BNF) and sustainable agriculture. Her research has significantly improved crop productivity while reducing dependence on chemical fertilizers, particularly in soybean cultivation in Brazil, making agriculture more environmentally sustainable and cost-effective.

Q 47. Ans. B

Explanation:

- India’s first fleet of hydrogen fuel cell buses was recently launched in Leh, Ladakh. The initiative is part of India’s push towards clean and green

hydrogen-based mobility, especially in high-altitude and environmentally sensitive regions.

- These hydrogen fuel cell buses generate electricity through a chemical reaction between hydrogen and oxygen, producing only water vapour as emission, making them a zero-emission transport solution.

Q 48. Ans. B

Explanation:

- The Diu district (Union Territory of Dadra and Nagar Haveli and Daman and Diu) became the first district in India to meet 100% of its daytime electricity demand through solar energy. This milestone was achieved through a combination of large-scale solar parks, rooftop solar installations, and solar net metering systems.
- Diu’s achievement is significant because it has transformed a region that was earlier power-deficit and dependent on electricity imports into a solar power surplus model, showcasing India’s renewable energy transition.

Q 49. Ans. C

Explanation:

- India’s first comics library at a government school setup was launched in Rajapalayam, Virudhunagar district, Tamil Nadu, in February 2025, established through a partnership between the Virudhunagar district administration and the comiclibrary.com.
- The Guwahati Gateway Terminal, inaugurated on November 7, 2025, by Union Finance Minister Nirmala Sitharaman on the southern bank of the Brahmaputra at Fancy Bazar, is recognized as India’s first modern river ferry terminal.
- India’s first Semiconductor Innovation Museum, a flagship initiative of T-CHIP (Technology Chip Innovation Program), was inaugurated in Hyderabad on October 12, 2025; while broader reports place it in Hyderabad generally, this aligns with the Kondapur location cited (Kondapur is a well-known IT/tech hub locality within Hyderabad).

Q 50. Ans. C

Explanation:

- **ZAPAD 2025:** An Indian Armed Forces contingent (primarily Army, with personnel from the Air Force and Navy too) participated in this multilateral joint military exercise hosted by Russia at the Mulino Training Ground, aimed at enhancing military cooperation and interoperability between India and Russia.

- **Tiger Claw:** This was the first-ever joint special operations exercise between the Garud Special Forces of the Indian Air Force and the United States Air Force, conducted from May 26 to June 10, 2025, correctly matching IAF and USAF.
- **Freedom Edge:** This is a trilateral multi-domain exercise involving South Korea, the United States, and Japan, conducted in international waters near Jeju Island to strengthen three-way security cooperation against North Korea's military threats, correctly matching the listed nations.

Q 51. Ans. C

Explanation:

- Russia — Order of St. Andrew the Apostle
 - ✓ Russia conferred its highest civilian honour, the Order of St. Andrew the Apostle, on PM Narendra Modi in recognition of his contribution to strengthening India–Russia strategic partnership.
- Brazil — Grand Collar of the National Order of the Southern Cross
 - ✓ Brazil awarded PM Modi the Grand Collar of the National Order of the Southern Cross, its highest civilian award given to foreign dignitaries, for enhancing bilateral relations.
- Egypt — Order of the Nile
 - ✓ Egypt conferred the Order of the Nile, its highest state honour, on PM Modi during his visit, recognising deepening India–Egypt strategic ties.
- France — Grand Cross of the Legion of Honour
 - ✓ France awarded PM Modi the Grand Cross of the Legion of Honour, its highest civilian decoration.

Q 52. Ans. C

Explanation:

- **Statement 1 is correct:** The 6th BIMSTEC Summit was hosted by Thailand, which currently serves as the BIMSTEC Chair. The summit brought together leaders from the Bay of Bengal Initiative for Multi-Sectoral Technical and Economic Cooperation (BIMSTEC) member countries to strengthen regional cooperation in areas such as trade, security, connectivity, and disaster management.
- **Statement 2 is correct:** The theme of the 6th BIMSTEC Summit was “BIMSTEC: Prosperous, Resilient and Open.” It reflects the collective vision of member states to promote economic prosperity, regional resilience, and greater openness in cooperation and integration.

Q 53. Ans. D

Explanation:

- **Statement 1 is correct:** Bihar became the first state to use mobile phone-based voting in municipal elections as a pilot initiative under the State Election Commission, aimed at improving accessibility and voter participation, especially for urban migrants and elderly voters.
- **Statement 2 is correct:** Himachal Pradesh became the first state to implement Aadhaar-based face authentication for Public Distribution System (PDS) ration distribution.
- This system enhances transparency, reduces fraud, and enables biometric verification even when fingerprints fail (e.g., elderly or manual labourers).
- **Statement 3 is correct:** Sikkim became the first Indian state to introduce a formal Sabbatical Leave Scheme for government employees, allowing long-term leave for personal development, higher studies, research, or caregiving without job loss.

Q 54. Ans. A

Explanation:

- India's first indigenous CRISPR-based gene therapy for Sickle Cell Disease has been developed under the initiative BIRSA 101.
- It represents a major breakthrough in gene editing technology (CRISPR-Cas system) aimed at correcting the defective gene responsible for sickle cell anaemia, a hereditary blood disorder.
- BIRSA 101 is associated with indigenous CRISPR-based therapeutic development in India.
- Targets Sickle Cell Disease (SCD), which is caused by a mutation in the HBB gene leading to abnormal haemoglobin (HbS).
- Aims to provide a potential one-time curative gene-editing treatment, especially relevant for tribal and high-prevalence regions in India.

Q 55. Ans. B

Explanation:

- MantraSana is the world's first portable robotic telesurgery system developed entirely in India, launched by SS Innovations International. It is designed to enable robot-assisted surgeries that can be performed remotely (telesurgery), significantly improving access to advanced surgical care in remote and underserved regions.
- The system is part of India's growing capabilities in medical robotics and digital healthcare innovation,

allowing surgeons to operate with precision using robotic arms controlled from a distant location through high-speed communication networks.

Q 56. Ans. C

Explanation:

- The Cold Desert Biosphere Reserve is located in Himachal Pradesh, in the trans-Himalayan region of Lahaul–Spiti and adjoining areas. It has recently been included in UNESCO's World Network of Biosphere Reserves (WNBR), highlighting its global ecological significance.
- This biosphere reserve is characterised by a cold desert ecosystem, with very low precipitation, high altitude, sparse vegetation, and extreme climatic conditions. Despite its harsh environment, it supports unique flora and fauna adapted to cold desert conditions.

Q 57. Ans. A

Explanation:

- **Assertion (A) is true:** The Minamata Convention on Mercury, adopted in 2013 and in force since 2017, is the global treaty aimed at protecting human health and the environment from the adverse effects of mercury and mercury compounds. At the Sixth Conference of the Parties (COP-6), held in Geneva from 3 to 7 November 2025, the 153 parties to the Convention took a landmark decision to phase out mercury-based dental amalgam. This represents a clear global commitment to eliminating dental amalgam use.
- **Reason (R) is true:** At COP-6, the parties adopted amendments to Annex A of the Convention, establishing a global phase-out of dental amalgam by 2034. After this date, the manufacture, import, and export of mercury-containing dental amalgam will no longer be permitted. The decision was reached after years of negotiation and was described by many delegations as a historic milestone for the Convention.
- **Why R explains A:** The 2034 phase-out decision is itself the substantive content of the commitment described in the Assertion. The Minamata Convention's commitment to eliminating dental amalgam use is given concrete shape precisely by the time-bound, science-based agreement to stop its manufacture and trade by 2034. Therefore, R is the direct and correct explanation of A.
- **Additional Facts:**
 - ✓ The Minamata Convention is named after the Japanese city of Minamata, where thousands of residents suffered severe mercury poisoning in

the 1950s after consuming fish contaminated with methylmercury from industrial wastewater. The disease came to be known as Minamata Disease.

- ✓ Dental amalgam, a mixture of liquid mercury with silver, tin, and copper, has been used as a tooth filling material for over 175 years. It is being replaced globally by mercury-free alternatives such as composite resins and glass ionomer cements.
- ✓ Around 50 countries, including all 27 European Union member states, had already phased out dental amalgam before COP-6. Many countries, including the United States, continue to allow its use.
- ✓ The Convention also addresses other major sources of mercury exposure, such as artisanal and small-scale gold mining (ASGM), mercury-containing products like batteries and thermometers, and emissions from coal-fired power plants and cement production.
- ✓ The Seventh Conference of the Parties (COP-7) is scheduled for 14-18 June 2027 in Geneva, coinciding with the tenth anniversary of the Convention's entry into force.

Q 58. Ans. B

Explanation:

- India has recently signed an Inter-Governmental Agreement (IGA) with France for the procurement of 26 Rafale-M (Marine) fighter aircraft specifically for the Indian Navy. These aircraft are naval variants of the Rafale fighter jet, designed for carrier-based operations.
- The Rafale-M will operate from India's aircraft carriers such as INS Vikrant and INS Vikramaditya, significantly enhancing the Navy's carrier-based combat capability, air defence, and maritime strike power.

Q 59. Ans. C

Explanation:

- Kirsty Coventry, the former Olympic swimming champion from Zimbabwe, has recently been elected as the first female and first African President of the International Olympic Committee (IOC). This marks a historic milestone in the governance of global sports administration.
- She is one of the most decorated African Olympians and has played a key role in sports administration and athlete representation before taking over as IOC President.

Q 60. Ans. A**Explanation:**

- Inamgaon, located in Maharashtra on the banks of the Ghod river, is the most extensively excavated Jorwe culture site and is known for evidence of mud fortification/embankment along with a large granary structure, interpreted by archaeologists as a chief's storage facility for surplus grain pointing to social stratification within the settlement. This makes it distinct from sites like Ahar and Gilund, which belong to the Ahar-Banas culture of Rajasthan, and Kayatha, associated with the Kayatha culture in Madhya Pradesh.

Q 61. Ans. C**Explanation:**

- Kalibangan — Evidence of earliest ploughed agricultural field
 - ✓ Kalibangan (Rajasthan), situated on the banks of the Ghaggar River, has yielded the earliest known archaeologically preserved ploughed agricultural field. The field shows a criss-cross pattern of furrows, indicating the cultivation of two crops simultaneously (mixed cropping), making it one of the most significant discoveries related to Harappan agriculture.
- Banawali — Discovery of a high-quality terracotta toy model of a plough
 - ✓ Banawali (Haryana), located on the dried-up course of the Saraswati (Ghaggar) River, has yielded a terracotta toy model of a plough. The discovery, along with evidence from Kalibangan, provides valuable insight into the agricultural practices and farming technology of the Harappan civilisation.
- Daimabad — Southernmost outpost yielding an extensive cache of bronze sculptures
 - ✓ Daimabad (Ahmednagar district, Maharashtra) is regarded as the southernmost Harappan-related site. It is renowned for the discovery of the Daimabad Bronze Hoard, comprising large bronze sculptures, including a chariot with a driver, an elephant, a rhinoceros, and a buffalo.

Q 62. Ans. C**Explanation:**

- Red Ware was the most common and numerically dominant pottery during the Later Vedic period (c. 1000–600 BCE). It was used extensively for everyday

domestic purposes such as cooking, storage, and serving food. Most settlements of the Later Vedic age have yielded large quantities of plain and coarse Red Ware, reflecting its widespread use among the general population.

- While Painted Grey Ware (PGW) is closely associated with the Later Vedic culture, it constituted only a small proportion of the total pottery assemblage. PGW was a fine grey pottery decorated with black geometric designs and is generally linked with the Kuru-Panchala region and the emergence of early Iron Age settlements. In contrast, Northern Black Polished Ware (NBPW) belongs to the later period (c. 700–200 BCE), associated with the Second Urbanisation and the Mahajanapada-Mauryan era. Black and Red Ware (BRW) continued at some sites but was not the dominant ceramic type of the Later Vedic period.

Q 63. Ans. A**Explanation:**

- **Statement 1 is correct:** Ancient punch-marked coins (6th century BCE onwards), the earliest metallic currency of India, did not bear alphabetic inscriptions, legends, dates, or the names of issuing rulers. Instead, they were identified by a series of symbols punched onto the metal surface, such as the sun, six-armed symbol, animals, trees, hills, and geometric motifs. These coins were issued by Mahajanapadas, merchant guilds, local authorities, and later by the Mauryan state.
- **Statement 2 is incorrect:** Silver, not gold, was the principal metal used in the manufacture of pre-Mauryan punch-marked coins. The standard silver coin was known as the Karshapana (Pana) and formed the backbone of the monetary system during the Mahajanapada period. Although a limited number of copper punch-marked coins are also known, gold was not the principal metal used for pre-Mauryan punch-marked coinage.

Q 64. Ans. A**Explanation:**

- Mudrarakshasa — Vishakhadatta
 - ✓ Mudrarakshasa is a Sanskrit political drama authored by Vishakhadatta. It narrates the political intrigues surrounding Chanakya's efforts to secure the throne of Chandragupta Maurya by winning over Rakshasa, the former minister of the Nanda dynasty.
- Malavikagnimitram — Kalidasa

- ✓ Malavikagnimitram is the earliest known play of Kalidasa. It revolves around the love story of King Agnimitra of the Shunga dynasty and the maiden Malavika. The play also provides valuable historical information about the Shunga rulers. He also authored Abhijnanashakuntalam, Vikramorvashiyam, Kumarasambhavam, Raghuvamsham, and Meghadutam.
- Mrichchhakatika — Shudraka
 - ✓ Mrichchhakatika (The Little Clay Cart) is a celebrated Sanskrit drama traditionally attributed to King Shudraka. The play depicts urban life, trade, social customs, and the lives of merchants and common people, making it an important source for understanding ancient Indian society.
- Buddhacharita — Ashvaghosha
 - ✓ Buddhacharita is an epic Sanskrit poem composed by Ashvaghosha, one of the greatest Buddhist scholars and poets. It presents a detailed account of the life of Gautama Buddha, from his birth to his enlightenment, and is regarded as one of the earliest classical Sanskrit mahakavyas.

Q 65. Ans. A

Explanation:

- The Aihole Inscription was composed by Ravikirti, the celebrated court poet of Pulakeshin II of the Early (Badami) Chalukya dynasty. Engraved in 634 CE at the Meguti Jain Temple in Aihole (Karnataka), the inscription is written in Sanskrit using the Kannada script. It is one of the most important epigraphic sources for the political of early medieval India.
- The inscription glorifies the achievements of Pulakeshin II, most notably his victory over Emperor Harshavardhana on the banks of the Narmada River, which established the Narmada as the boundary between the Chalukya kingdom in the south and Harsha's empire in the north.

Q 66. Ans. A

Explanation:

- **Assertion (A) is true:** The Chinese Buddhist pilgrim Fa-Hien (Faxian), who visited India during the reign of Chandragupta II (c. 399–414 CE), recorded that Chandalas lived outside the main settlements. According to his account, when they entered towns or marketplaces, they were required to strike a piece of wood to warn others of their approach so

that people could avoid physical contact with them.

- **Reason (R) is true:** Untouchability became more pronounced during the Gupta period than in earlier times. Literary sources such as the Dharmashastras and Fa-Hien's account indicate increasing social stratification and stricter observance of caste norms. Groups such as the Chandalas, associated with occupations considered ritually impure, were subjected to social exclusion and residential segregation.

Q 67. Ans. C

Explanation:

- **Statement 1 is correct:** Pushyamitra Shunga, the founder of the Shunga dynasty (c. 185 BCE), revived orthodox Brahmanical traditions after overthrowing the last Mauryan ruler, Brihadratha. Literary sources, particularly the Ayodhya Inscription of Dhanadeva and Patanjali's Mahabhashya, indicate that he performed Ashvamedha (horse) sacrifices to proclaim his imperial authority and political supremacy.
- **Statement 2 is correct:** The stone railings and gateways surrounding the Great Stupa at Sanchi were constructed and enlarged during the Shunga period. Although the original brick stupa was built under Emperor Ashoka, the Shungas enlarged the stupa, encased it with stone, and added the stone balustrades (railings), contributing significantly to the development of early Buddhist architecture.

Q 68. Ans. B

Explanation:

- The Kushana rulers, particularly Kanishka, adopted the title "Devaputra" (Son of Heaven/God) as part of their elaborate royal titlature to assert divine kingship. This concept was likely borrowed from Chinese political tradition, where the emperor was referred to as the "Son of Heaven" (Tianzi), reflecting the Kushanas' cosmopolitan empire that straddled Central Asia and had contact with Han China.
- Maharajadhiraja ("Great King of Kings") was a Sanskrit title used by various Indian dynasties, including the Kushanas themselves alongside Devaputra, but it doesn't specifically translate to "Son of Heaven." Shahanshah is the Persian-origin title meaning "King of Kings," and Kaiser/Kaisara is the Kushana adaptation of the Roman imperial title "Caesar" — both were part of Kushana royal titles too, but neither carries the specific meaning of "Son of Heaven."

Q 69. Ans. D**Explanation:**

- **Statement 1 is incorrect:** The Cartaz system was not egalitarian, nor did it promote open transit. It was a coercive naval trade-licensing system under which any merchant ship passing through the Indian Ocean had to buy a pass from the Portuguese. If a ship failed to produce a Cartaz, the Portuguese Navy would routinely seize the vessel, confiscate the cargo, and execute or enslave the crew.
- **Statement 2 is incorrect:** The strategic architecture of the Estado da Índia relied heavily on coastal naval checkpoints and maritime choke points (like Goa, Malacca, and Ormuz) to enforce its commercial monopoly, rather than deep inland territorial annexation. They controlled the sea lanes using naval supremacy rather than occupying large expanses of land.

Q 70. Ans. C

- The Paika Rebellion of 1817 in Odisha was one of the earliest armed uprisings against British rule. The rebellion was led by Bakshi Jagabandhu, the hereditary military chief (Bakshi) of the Khurda kingdom.
- **The primary causes were:**
 - ✓ The British conquest and annexation of Khurda (1803).
 - ✓ Withdrawal of rent-free land grants (Nishkar lands) traditionally enjoyed by the Paikas.
 - ✓ Heavy and oppressive land revenue demands.
 - ✓ Economic hardship faced by peasants, zamindars, and Paikas.
 - ✓ Displacement of the traditional military-landholding class under British administration.
- The Paikas, who had traditionally rendered military service in return for land grants, lost their economic base due to British revenue policies. This directly triggered widespread resentment and rebellion.

Q 71. Ans. A**Explanation:**

- The Ghalzai (Ghilzai) tribesmen, led by Mir Wais Hotak and later by his son Mahmud Hotak, successfully detached Qandahar from the Safavid Empire in the early eighteenth century. In 1721–22, Mahmud Hotak launched an invasion of Persia and, after defeating the Safavid forces at the Battle of Gulnabad (1722), captured the Safavid capital, Isfahan, bringing an end to the effective rule of the Safavid dynasty.

- The conquest of Isfahan marked the temporary establishment of the Hotak dynasty over much of Persia. However, the Ghalzai rule was short-lived, as Nadir Shah Afshar eventually defeated the Hotaks and restored Persian authority.

Q 72. Ans. D**Explanation:**

- Nadir Shah signed the Treaty of Ganja (1735) with Russia, under which Russia agreed to restore the Caspian provinces it had occupied during the decline of the Safavid Empire. After consolidating his military and political power, Nadir Shah deposed the last Safavid ruler, Abbas III, in 1736 and established the Afsharid dynasty, becoming the Shah of Persia.
- In 1738–39, Nadir Shah invaded the north-western frontier of India, exploiting the weakening Mughal Empire. He decisively defeated Emperor Muhammad Shah at the Battle of Karnal (1739) and subsequently occupied Delhi, carrying away immense wealth, including the Peacock Throne and the Koh-i-Noor diamond.

Q 73. Ans. B**Explanation:**

- **Statement 1 is correct:** Nizam-ul-Mulk (Asaf Jah I) played a leading role in the overthrow of the Saiyid brothers, who had dominated Mughal politics as the “Kingmakers” after the death of Aurangzeb. He supported Emperor Muhammad Shah in ending the political influence of the Saiyid brothers, culminating in the fall of Saiyid Husain Ali Khan (1720) and Saiyid Abdullah Khan (1720).
- **Statement 2 is incorrect:** Although Nizam-ul-Mulk established his authority over Hyderabad in 1724 after defeating Mubarez Khan at the Battle of Shakar Kheda, he did not openly declare independence from the Mughal Empire. He continued to acknowledge the nominal suzerainty of the Mughal emperor, issued coins and read the khutba in the emperor’s name, while exercising virtual autonomy in the Deccan.

Q 74. Ans. C**Explanation:**

- **Statement 1 is correct:** Baji Rao I (1720–1740) successfully drove the Sidis of Janjira out of most of their mainland possessions, though the fortified island of Janjira itself remained under Sidi control. This strengthened Maratha authority along the

Konkan coast and reduced the Sidis' influence in the region.

- **Statement 2 is incorrect:** Before the Third Battle of Panipat (1761), the Marathas failed to secure durable alliances with several important north Indian powers, including the Rajputs and Jat ruler Suraj Mal. Differences in strategy and political disagreements led these potential allies to withhold military support, leaving the Marathas relatively isolated against Ahmad Shah Abdali.
- **Statement 3 is correct:** The Maratha army at the Third Battle of Panipat included a well-trained European-style infantry commanded by Ibrahim Khan Gardi. His troops were equipped with modern muskets and field artillery, trained on French military lines, and initially inflicted heavy casualties on Abdali's forces before the Maratha defeat.

Q 75. Ans. B

Explanation:

- In 1623, the Dutch East India Company (VOC) decisively thwarted English efforts to establish themselves in the Indonesian Islands following the infamous Amboyna (Amboina) Incident. At Amboyna in the Moluccas (Spice Islands), the Dutch arrested, tortured, and executed several English East India Company officials on charges of conspiracy.
- Following this setback, the English East India Company gradually shifted its commercial focus to the Indian subcontinent, where it expanded its trading settlements at Surat, Madras, Bombay, and Calcutta, laying the foundation for British political power in India.

Q 76. Ans. B

Explanation:

- The English East India Company established its first factory in southern India at Masulipatam (Machilipatnam) in 1611. Located on the Coromandel Coast in present-day Andhra Pradesh, Masulipatam was an important port under the Kingdom of Golconda and emerged as the Company's earliest trading base in South India. From here, the English carried on a flourishing trade in cotton textiles, dyed fabrics, indigo, and spices.
- Subsequently, in 1639, the Company acquired Madraspatnam, where Fort St. George was built, and Madras (Chennai) developed into the principal English settlement in South India.

Q 77. Ans. B

Explanation:

- Must be truthful
 - ✓ Truth (Satya) formed the very foundation of Satyagraha. According to Mahatma Gandhi, a Satyagrahi must be absolutely truthful in thought, word, and action. Gandhi regarded Truth and Non-violence (Ahimsa) as inseparable principles and believed that adherence to truth was essential for achieving justice through moral means.
- Must willingly suffer while resisting injustice
 - ✓ A true Satyagrahi must be prepared to accept suffering voluntarily rather than inflict suffering on others. Gandhi maintained that self-suffering had the power to awaken the conscience of the oppressor and bring about social and political change without the use of violence. This principle distinguished Satyagraha from passive resistance.
- Must hate the wrongdoer but oppose the wrong
 - ✓ This statement is incorrect. Gandhi consistently emphasized the principle of "Hate the sin, not the sinner." A Satyagrahi must oppose injustice and evil, but should never harbour hatred towards the person committing the wrong.

Q 78. Ans. A

Explanation:

- Gandhiji went to South Africa to practise law (1893)
 - ✓ Mahatma Gandhi travelled to South Africa in 1893 as a legal representative for an Indian merchant. His experiences of racial discrimination there, particularly the Pietermaritzburg railway incident, profoundly influenced his political philosophy and laid the foundation for his future methods of resistance.
- Evolution of Satyagraha in South Africa (1906)
 - ✓ The concept of Satyagraha evolved in 1906 during Gandhi's struggle against the Asiatic Registration Act (Black Act) in the Transvaal. It marked the beginning of a new method of non-violent resistance based on truth (Satya) and non-violence (Ahimsa).
- Gandhiji returned to India (1915)
 - ✓ After spending over two decades in South Africa, Gandhi returned to India on 9 January 1915 at the invitation of Gopal Krishna Gokhale. The day is now commemorated as Pravasi Bharatiya Divas.

- Foundation of Sabarmati Ashram (1917)
 - ✓ Gandhi established the Sabarmati Ashram at Ahmedabad in 1917 on the banks of the Sabarmati River. It became the headquarters of his national movements, including the Non-Cooperation Movement and the Salt (Dandi) March.

Q 79. Ans. C

Explanation:

- Vallabhbhai Patel emerged as one of Mahatma Gandhi's closest associates during the Kheda Satyagraha (1918) in Gujarat. The movement was launched to demand the suspension of land revenue collection after crop failure caused by drought. Patel played a crucial role in organising the peasants, mobilising village-level support, coordinating volunteers, and maintaining discipline throughout the campaign.
- The successful conclusion of the Kheda Satyagraha established Vallabhbhai Patel as a prominent nationalist leader and marked the beginning of his long association with Gandhi. Patel later led the Bardoli Satyagraha (1928), after which he earned the title "Sardar" from the peasants.

Q 80. Ans. C

Explanation:

- **The 1898 amendment to the Indian Penal Code** (Section 124A) broadened the definition of sedition, making it easier to prosecute anyone deemed to be inciting "disaffection" against the government, severely curbing free political expression and press criticism.
- **The Indian Official Secrets Act (1904)** restricted access to official information and was seen as a tool to suppress journalism and stifle scrutiny of government actions, limiting press freedom.
- **The Indian Universities Act (1904)** tightened government control over universities by increasing official representation on governing bodies and senates, which nationalists viewed as an attempt to curb the growth of independent, nationalist-minded intellectual and student activity.
- The Indian Councils Act of 1892, by contrast, was not viewed as a restrictive or repressive measure. It actually expanded the size of legislative councils and introduced limited indirect elections, along with allowing members to discuss the budget and ask questions.

Q 81. Ans. B

Explanation:

- Swami Vivekananda made the statement, "The only hope of India is from the masses." He believed that India's regeneration depended not on a small educated elite but on the awakening, education, and upliftment of the common people, especially the poor and the downtrodden. He repeatedly emphasized that true nation-building required empowering the masses through education, self-confidence, character-building, and service to humanity.

Q 82. Ans. D

Explanation:

- **Statement 1 is correct:** The Partition of Bengal (1905) created the new province of Eastern Bengal and Assam, with Dacca (Dhaka) as its capital. This new province had a Muslim majority population, comprising eastern Bengal and Assam.
- **Statement 2 is correct:** The remaining western province of Bengal, with Calcutta as its capital, retained western Bengal, Bihar, and Orissa and had a Hindu majority. The reorganisation thus divided Bengal into two provinces with distinct religious demographics.
- **Statement 3 is correct:** Although Lord Curzon justified the partition on grounds of administrative convenience, Indian nationalists regarded it as a classic example of the British policy of "Divide and Rule." They believed that the partition deliberately divided the Bengali-speaking population on territorial as well as religious lines to weaken the growing nationalist movement centred in Bengal.

Q 83. Ans. B

Explanation:

- The Hindu Succession Act, 1956 was enacted to codify and reform the law relating to intestate succession among Hindus. It significantly improved the property rights of women by recognizing daughters as Class I heirs, placing them in the same category as sons, widows, and mothers for inheritance in the separate property of a deceased Hindu dying intestate.
- However, under the original Act of 1956, daughters were not equal coparceners in a Mitakshara Joint Hindu Family. This inequality was removed through the Hindu Succession (Amendment) Act, 2005, which granted daughters the same coparcenary rights and liabilities as sons by birth.

Q 84. Ans. A

Explanation:

- **Statement 1 is correct:** The Swadeshi Movement (1905–08) encouraged the establishment of Indian-owned banks, insurance companies, and commercial enterprises to reduce dependence on British capital. Institutions such as the Bengal National Bank, National Insurance Company, and several indigenous financial enterprises emerged during this period, reflecting the objective of achieving economic self-reliance.
- **Statement 2 is correct:** The movement actively promoted the establishment and expansion of indigenous industries, including textile mills, soap factories, match factories, tanneries, handloom units, and other small-scale industries. The use of Swadeshi goods and the boycott of foreign products became central features of the movement's economic programme.
- **Statement 3 is incorrect:** The Swadeshi Movement did not discourage chemical industries. On the contrary, nationalists strongly supported the development of modern indigenous industries, including chemical manufacturing. A notable example was the Bengal Chemical and Pharmaceutical Works, founded by Acharya Prafulla Chandra Ray, which became a symbol of India's industrial self-reliance.

Q 85. Ans. D

Explanation:

- Acharya P.C. Ray — Bengal Chemical Swadeshi Stores
 - ✓ Acharya Prafulla Chandra Ray was an advocate of Swadeshi industries. He founded the Bengal Chemical and Pharmaceutical Works in 1901, India's first successful indigenous chemical and pharmaceutical enterprise. During the Swadeshi Movement, he actively promoted Bengal Chemical Swadeshi Stores to encourage the production and consumption of Indian-made goods.
- Aurobindo Ghose — National College, Calcutta
 - ✓ Aurobindo Ghose served as the first Principal of the National College, Calcutta, established to provide national education free from colonial control. Through his writings and teaching, he inspired students to participate in the nationalist movement and promoted the ideals of self-reliance and national awakening.

- Rabindranath Tagore — Helped establish a Swadeshi Store

- ✓ Rabindranath Tagore actively supported the Swadeshi Movement by encouraging the use of indigenous goods and promoting economic self-reliance. He was associated with the establishment of Swadeshi stores and organized Raksha Bandhan ceremonies to symbolize the unity of Hindus and Muslims against the Partition of Bengal (1905).

- National Council of Education — Established on 15 August 1906

- ✓ The National Council of Education (NCE) was established on 15 August 1906 in Bengal to develop a national system of education independent of British control. It later laid the foundation for the establishment of Jadavpur University, becoming an important institution of the national education movement.

Q 86. Ans. A

Explanation:

- **Statement 1 is correct:** The nationalist press played a vital role in exposing and criticising colonial administrative policies such as excessive taxation, economic exploitation, racial discrimination, and restrictions on civil liberties. It articulated the Indian point of view, countered official propaganda, and became an important instrument for mobilising public opinion against British rule.
- **Statement 2 is correct:** Newspapers and journals were instrumental in spreading modern political ideas such as self-government (Swaraj), democracy, civil rights, representative institutions, national unity, and industrialisation. They also educated the public on issues like the drain of wealth, economic nationalism, Swadeshi, and the need for indigenous industries.
- **Statement 3 is incorrect:** Far from discouraging communication, the nationalist press strengthened contacts among nationalist workers across different regions of India. Newspapers enabled the rapid exchange of ideas, coordinated political campaigns, and fostered a sense of all-India nationalism, helping transform local grievances into a national movement.

Q 87. Ans. D

Explanation:

- Bankim Chandra Chatterjee — Bengali
 - ✓ Bankim Chandra Chatterjee was one of the greatest literary figures of the Bengali Renaissance. He wrote predominantly in

Bengali, and his novel Anandamath contains the patriotic song “Vande Mataram,” which became a powerful symbol of the Indian nationalist movement.

- Lakshminath Bezbarua — Assamese
 - ✓ Lakshminath Bezbarua was a pioneering Assamese writer, poet, playwright, and journalist. A leading figure of the Jonaki Era in Assamese literature, he played a significant role in the revival and modernisation of the Assamese language and literature.
- Vishnu Shastri Chiplunkar — Marathi
 - ✓ Vishnu Shastri Chiplunkar was a renowned Marathi writer, literary critic, and nationalist. Through his influential journal Nibandhamala, he promoted social reform, national consciousness, and the development of modern Marathi prose.
- Altaf Husain Hali — Urdu
 - ✓ Altaf Husain Hali was one of the foremost poets and literary critics of Urdu literature. His celebrated work Musaddas-e-Hali (Madd-o-Jazr-e-Islam) emphasized social and educational reform among Indian Muslims and greatly influenced the Aligarh Movement.

Q 88. Ans. D

Explanation:

- 1860–61 — Western Uttar Pradesh
 - ✓ The famine of 1860–61 primarily affected Western Uttar Pradesh, particularly the Upper Doab region, along with parts of the Punjab. It exposed the inadequacy of colonial famine relief measures and highlighted the vulnerability of the agrarian economy under British revenue policies.
- 1865–66 — Orissa
 - ✓ The Orissa Famine of 1865–66 was one of the most devastating famines of the nineteenth century. It caused the death of nearly one-third of the population of Orissa.
- 1876–78 — Mysore
 - ✓ The Great Famine of 1876–78 affected large parts of Madras Presidency, Mysore, Hyderabad, Bombay Presidency, and the Central Provinces. Mysore was among the worst-affected regions. The famine led to millions of deaths and resulted in the appointment of the Famine Commission (1878) under Sir Richard Strachey, whose recommendations formed the basis of the Famine Code.

● 1943 — Bengal

- ✓ The Bengal Famine of 1943 occurred during the Second World War and caused the death of an estimated 2–3 million people. A combination of wartime inflation, disruption of rice supplies, administrative failures, and colonial procurement policies aggravated the crisis. The famine became one of the strongest indictments of British rule in India.

Q 89. Ans. B

Explanation:

- After the Government of India Act, 1858, the British Parliament remained the supreme authority over Indian affairs. The Act abolished the East India Company and transferred the administration of India directly to the British Crown. The Secretary of State for India, a member of the British Cabinet, exercised extensive control over Indian administration and was responsible to Parliament.
- Although the Governor-General (Viceroy) administered India on behalf of the Crown and the Council of India advised the Secretary of State, both functioned under the ultimate authority of the British Parliament.

Q 90. Ans. D

Explanation:

- Giuseppe Mazzini — Inspired Indian Nationalists
 - ✓ A leading Italian nationalist and founder of Young Italy, Mazzini’s ideas of national unity, patriotism, and sacrifice deeply influenced Indian leaders such as Surendranath Banerjea, Bipin Chandra Pal, Lala Lajpat Rai, and Aurobindo Ghose.
- Giuseppe Garibaldi — Symbol of Revolutionary Nationalism
 - ✓ Garibaldi played a key role in the unification of Italy through his military campaigns. His courage and commitment to national liberation inspired Indian nationalists in their struggle against colonial rule.
- Irish Nationalist Leaders — Inspired the Home Rule Movement
 - ✓ The Irish Home Rule Movement influenced Indian nationalists, who admired its constitutional and mass-based methods. The Home Rule Movement in India, led by Bal Gangadhar Tilak and Annie Besant, drew inspiration from the Irish experience.

Q 91. Ans. B

Explanation:

- **Kautilya (Chanakya)** is associated with **Takshashila**, not Nalanda, and lived much earlier than Nalanda's peak period.
- He served as a chief advisor to **Chandragupta Maurya**, founder of the Mauryan Empire.
- He studied and taught at **Taxila (Takshashila)**, which was a famous center of learning.
- The **Arthashastra** remains important because it gives detailed insight into ancient Indian ideas about administration, taxation, law, and foreign policy.
- **Nagarjuna** is widely regarded as an important Buddhist philosopher connected with the Nalanda intellectual tradition. He is the founder of the Madhyamaka (Middle Way) school of Mahayana Buddhism
- **Sheelbhadra** was a famous head/teacher at Nalanda in the 7th century and taught Xuanzang.
- **Shantirakshit** was also a renowned Nalanda scholar who later helped establish Buddhism in Tibet.

Q 92. Ans. C

Explanation:

- **National Planning Committee (1938):-** The idea of planned economic development in India was first formally initiated in 1938 by the then President of the Indian National Congress, **Netaji Subhas Chandra Bose**. He constituted the **National Planning Committee (NPC)**, with **Jawaharlal Nehru** as its Chairman. The committee comprised **15 members** and emphasized that **national independence was a prerequisite** for implementing a comprehensive economic plan.
- The **Sarvodaya Plan** was drafted by **Jayaprakash Narayan** in 1950. Inspired by **Mahatma Gandhi's** economic philosophy and **Vinoba Bhave's** Sarvodaya movement, the plan emphasized agriculture, small-scale and cottage industries, land reforms, and decentralized participatory planning. It also advocated reducing dependence on foreign technology and promoting self-reliant rural development.
- The **People's Plan** was formulated in 1945 by **M.N. Roy** as a **ten-year development plan**. Unlike the Bombay Plan, it emphasized **collectivization of agriculture**, **state ownership of key industries**, and the development of **agriculture and consumer goods industries**. The plan advocated a socialist approach to economic development.

- In 1944, a group of eight leading industrialists and economists, including **Sir Purshottamdas Thakurdas** and **J.R.D. Tata**, proposed the **Bombay Plan**. The plan aimed to stimulate debate on India's economic development and provide guiding principles for a future national plan. It sought to **double the per capita income within 15 years** and proposed an increase of approximately **130% in agricultural production** and **500% in industrial production**.

Q 93. Ans. B

Explanation:

- The **notification issued on 28 January 1916** marked a significant administrative milestone in the development of Patna as the provincial capital. While the **Province of Bihar and Orissa** had been created in 1912, the 1916 notification formally brought Patna's role as the capital into full administrative operation after the completion of key government infrastructure.
- Issued under the administration of **Sir Charles Stuart Bayley**, the notification officially recognized **Patna** as the functioning capital of the province, aligning its legal and administrative status with the newly completed Secretariat, Government House, and other public buildings.
- The notification also introduced an important **municipal reorganization**. The existing **Patna Municipality**, established in 1864, was redesignated as the **Patna City Municipality**, paving the way for a separate municipal administration to govern the rapidly expanding capital area and meet its growing civic and administrative needs.
- Shortly thereafter, on **3 February 1916**, the Viceroy of India, **Lord Hardinge**, visited Patna to formally inaugurate the newly constructed administrative complex, marking the commencement of the city's full-fledged functioning as the provincial capital.

Q 94. Ans. A

Explanation:

- The foundation stone of **Bihar Lok Bhavan** (formerly known as **Government House** or **Raj Bhavan**) was laid by **Lord Hardinge (Lord Charles Baron Hardinge)**, the then Viceroy and Governor-General of India, on **1 December 1913**.
- On the same occasion, Lord Hardinge also laid the foundation stones for the three principal administrative buildings of the new provincial capital—**Government House (now Bihar Lok Bhavan)**, the **Old Secretariat**, and the **Patna High**

Court—forming the core of Patna's administrative complex.

- The architectural design of Bihar Lok Bhavan was prepared by **Joseph Fearis Munnings**, a New Zealand-born architect commissioned by the colonial government to plan several of the major public and administrative buildings in Patna's newly developed capital area.

Q 95. Ans. B

Explanation:

- With the creation of the **Province of Bihar and Orissa on 22 March 1912**, the British Government established a separate provincial administration headed by **Sir Charles Stuart Bayley**. To assist the Lieutenant Governor in the administration of the newly formed province, an **Executive Council** comprising **three members** was constituted.
 - ✓ **The three members of the Executive Council were:**
 - ✓ **Rameshwar Singh**
 - ✓ **Edward Albert Gait**
 - ✓ **Edward Vere Levinge**
- The inclusion of **Maharaja Rameshwar Singh** was particularly noteworthy, as he became one of the earliest prominent Indian representatives in the provincial executive administration. His appointment reflected the British policy of associating influential Indian princes and landlords with governance while retaining overall executive control in British hands.

Q 96. Ans. D

Explanation:

- **Statement 1 is correct:** The then Viceroy, Lord Hardinge, made the final formal proclamation regarding the creation of the **Province of Bihar and Orissa through the Government of India Act, 1912**.
- **Statement 2 is correct:** The proclamation was issued on **22 March 1912**, formally defining the **territorial boundaries** of the new province and bringing it into existence. With the establishment of the new province, the office of the **Lieutenant Governor in Council** was also constituted to administer the province. Later, on **1 April 1936**, Bihar and Orissa were separated into two distinct provinces. After this separation, the Province of Bihar comprised **four divisions**:
 - ✓ **Bhagalpur Division**
 - ✓ **Patna Division**
 - ✓ **Tirhut Division**
 - ✓ **Chota Nagpur Division**

Q 97. Ans. C

Explanation:

- **Pair 1 is correct:** **Madhubani Painting**, popularly known as **Mithila Art**, is a traditional folk art form that originated in the **Mithila region of Bihar**. It is distinguished by intricate line drawings filled with vibrant colours, bold contrasts, and elaborate geometric and floral patterns. Traditionally, these paintings were created by women on freshly plastered mud walls and floors of their homes using natural pigments. Over time, the art has evolved onto handmade paper, canvas, and cloth, with both men and women now practicing it to meet growing national and international demand. In recognition of its unique cultural heritage and regional identity, **Madhubani Painting was granted a Geographical Indication (GI) tag in 2007**.
- **Pair 2 is correct:** **Sujani** is a traditional embroidery craft that originated in **Bhusura village of Bihar**. Historically, women stitched together pieces of old cloth to create quilts for newborn babies and decorated them with intricate embroidery depicting deities, animals, plants, and scenes from everyday life. Over the years, Sujani has developed into a celebrated textile art known for its fine needlework and storytelling motifs. The craft received the **Geographical Indication (GI) tag in 2006**, recognizing its distinct regional identity. Further enhancing its international reputation, **Sujani Art was awarded the UNESCO Seal of Excellence (Seal Award) in 2019** for its outstanding craftsmanship and cultural significance.
- **Pair 3 is correct:** On **28 March 2018**, the **Bhagalpuri Zardalu Mango** was awarded the **Geographical Indication (GI) tag** by the Geographical Indications Registry, Chennai, under Registration No. 551. This premium mango variety is cultivated primarily in **Bhagalpur** and the adjoining districts of Bihar. Renowned for its creamy-yellow pulp, exceptional sweetness, rich flavour, and distinctive aroma, the Zardalu mango is considered one of Bihar's finest horticultural products. According to historical accounts, its cultivation in the region is believed to have been initiated by **Maharaja Rahmat Ali Khan Bahadur of Khadagpur Estate**, contributing to its enduring legacy in Bihar's agricultural heritage.

Q 98. Ans. B

Explanation:

- The inauguration of **PARAM Rudra**, Bihar's

first supercomputer, at IIT Patna represents a significant milestone in strengthening the state's scientific and technological infrastructure. With a computing capacity of approximately **838 teraflops**, this indigenously developed **High-Performance Computing (HPC)** system provides advanced computational resources for research in artificial intelligence, machine learning, data science, computational modelling, and other data-intensive fields across eastern India.

- PARAM Rudra is capable of substantially reducing the time required for complex scientific computations and large-scale simulations, thereby enhancing research efficiency and innovation. At IIT Patna, the supercomputer is expected to support around **60 faculty members** and **400 students** from **10 academic departments**, enabling cutting-edge research and interdisciplinary collaboration.

Q 99. Ans. A

Explanation:

- **Assertion (A)** is true Bihar has a **humid subtropical monsoon climate**, receiving nearly **80–85% of its annual rainfall from the Southwest Monsoon** between June and September. However, despite this strong monsoon influence, the state exhibits a pronounced **continental climate**, characterized by very **hot summers**, **cold winters**, and a **wide annual range of temperature**. This combination of monsoonal rainfall and continental temperature extremes is a distinctive feature of Bihar's climate.
- **Reason (R)** is true: Bihar is a **landlocked state** situated in the **middle Gangetic Plain**, far from the Arabian Sea and the Bay of Bengal. Since it is not influenced by the sea's moderating effect, the land heats up rapidly during summer and cools quickly during winter. As a result, the state experiences **high seasonal temperature variations**, with temperatures often rising above **40°C** in summer and falling below **5°C** in winter in some regions. This inland location is the primary reason for Bihar's **continentality**.
- The **Reason** directly explains the **Assertion**. Although Bihar receives abundant rainfall due to the **Southwest Monsoon**, its **distance from the sea** results in the absence of maritime moderation, producing a **continental climate** with significant seasonal temperature variations. Hence, both the **Assertion** and the **Reason** are true, and the **Reason** correctly explains the **Assertion**.

Q 100. Ans. C

Explanation:

- **Assertion (A)** is true The soils of Bihar are predominantly **alluvial**, deposited by the Ganga and its tributaries. Although these soils are highly fertile and suitable for crops such as rice, wheat, maize, sugarcane, and pulses, they are generally **deficient in nitrogen, phosphoric acid (phosphorus), and organic matter (humus)**.
- **Reason (R)** is false: While alluvial soils are naturally fertile and contain adequate amounts of minerals such as **potash and lime**, they are **not uniformly rich in all essential plant nutrients**. They are commonly deficient in **nitrogen, humus, and phosphorus**, and therefore require periodic application of fertilizers and organic manures.

Q 101. Ans. C

Explanation:

- **Assertion (A)** is true: During the **summer season (April to June)**, Bihar experiences **Loo**, which are **strong, hot, and dry winds** blowing mainly during the afternoon. These winds originate over the intensely heated northwestern plains of India and raise daytime temperatures significantly, often pushing them above **40°C**. The Loo is especially prominent in **south and central Bihar**, contributing to heatwave conditions and making summer the hottest season in the state.
- **Reason (R)** is false: Loo winds are **not cool or moisture-laden**. Instead, they are **extremely hot, dry, and desiccating winds** with very low humidity. Rather than reducing heat stress, they **intensify heat conditions**, causing severe discomfort and increasing the risk of **heatstroke, dehydration, and heat-related illnesses**. The arrival of the **Southwest Monsoon** eventually brings cooler, moisture-laden winds that end the Loo conditions.
- **Why is the Reason incorrect**
 - ✓ The Reason incorrectly describes the characteristics of the Loo. It is the **Southwest Monsoon winds**, not the Loo, that are cool (comparatively), moisture-bearing, and provide relief from the intense summer heat. Therefore, the Reason is false and does not explain the Assertion.

Q 102. Ans. B

Explanation:

- **Dr. Manish Mandal**, a senior surgeon at **Indira**

Gandhi Institute of Medical Sciences (IGIMS), Patna, has been elected as the **Governor of the Asia Pacific Region of the International College of Surgeons (ICS)**. The election took place during the **44th Biennial Congress of the ICS**, held in **Singapore on 5 December 2025**. He will serve a **two-year term** in this prestigious international position.

- The **International College of Surgeons (ICS)** is a global professional organization that brings together surgeons from across the world to promote excellence in **surgical education, research, training, and international collaboration**. Dr. Mandal's election marks a significant achievement for Bihar and highlights the growing global recognition of Indian medical professionals.

Q 103. Ans. C

Explanation:

- The **Chief Secretary of Bihar** felicitated **Anukul Roy** at **Vidyut Bhawan, Patna**, in recognition of his outstanding performance in the **2025–26 Syed Mushtaq Ali Trophy**. Roy was adjudged the **Player of the Tournament**, reflecting his exceptional contribution during the competition.
- Apart from his achievements in domestic cricket, Anukul Roy is associated with the **Bihar State Power Holding Company Limited (BSPHCL)**. His success has brought distinction not only to the sporting community but also to his organization and the State of Bihar.

Q 104. Ans. A

Explanation:

- The **Indrapuri Barrage**, popularly known as the **Indrapuri Dam** or **Sone Barrage**, is one of Bihar's most significant irrigation projects. Constructed across the **Sone River** near **Dehri-on-Sone** in **Rohtas district**, the barrage was completed in **1968**. The **Sone River**, one of the major southern tributaries of the **Ganga**, originates near **Amarkantak** in **Madhya Pradesh**, close to the source of the **Narmada River**. From its source, the river initially flows in a **northwestward** direction through **Madhya Pradesh** before taking a sharp **eastward** turn near the **Kaimur Range**. It then flows parallel to the **Kaimur Hills**, traversing the states of **Uttar Pradesh**, **Jharkhand**, and **Bihar**, before finally joining the **Ganga River** near **Maner** in **Patna district**.

Q 105. Ans. B

Explanation:

- **Bindeshwari Devi** was a renowned artisan from

Bihar and the most celebrated exponent of **Sikki grass craft**. Overcoming personal hardships, she played a pivotal role in preserving and promoting this traditional handicraft made from the golden **Sikki grass** found in Bihar's wetlands. Using intricate weaving techniques, she crafted baskets, boxes, toys, and decorative items that gained national and international recognition. In **1969**, she received the **National Award for Handicrafts**, presented by **Indira Gandhi**. Her work was also exhibited at several **Festival of India** events abroad, bringing global recognition to Bihar's **Sikki craft**.

Q 106. Ans. D

Explanation:

- **Statement 1 is correct:** **Manjusha Art** is one of Bihar's oldest folk painting traditions and is popularly known as **scroll painting**. It is regarded as one of the few Indian folk art forms that narrate a story through a **sequential series of paintings**, making it a distinctive narrative art. Originating in the **Ang region (Bhagalpur, Bihar)**, the art is believed to date back to the **7th century**, with some traditions tracing its roots even earlier.
- **Statement 2 is correct:** **Manjusha Art** is based on the folk legend of **Bihula-Bishari (Bihula-Bishahari)** and is closely associated with **Bishari Puja**, a Hindu festival dedicated to the serpent goddess **Bishahari (Manasa Devi)**. The paintings depict episodes from the **Bihula-Bishari legend**, making it a **story-oriented and subject-oriented** art form. It reflects the cultural and religious heritage of the **Ang region** of Bihar.
- **Statement 3 is correct:** In **2012**, the first **Sita Devi Award** in the field of **Manjusha Art** was **posthumously conferred on Late Chakravarty Devi** in recognition of her outstanding contribution to preserving and promoting this traditional folk art. Subsequently, in **2013**, **Shrimati Nirmala Devi** received the **Bihar Kala Award (Sita Devi Award)**.

Q 107. Ans. B

Explanation:

- The **Champaran Inquiry Committee** held its preliminary meeting on **11 July 1917**. After conducting several sittings and field inspections, it submitted its final report on **4 October 1917**. The Government accepted most of the Committee's recommendations, which significantly benefited the **ryats (tenant farmers)**. The most important recommendation was the **complete abolition of the**

Tinkathia system, under which cultivators were compelled to grow indigo on a portion of their land. Although the British indigo planters strongly opposed these reforms, they were unable to prevent the enactment of the **Champaran Agrarian Act** by the **Bihar and Orissa Legislative Council** on **4 March 1918**. The Act effectively ended the oppressive system of forced indigo cultivation, marking a historic victory for the Champaran Satyagraha.

Q 108. Ans. C

Explanation:

- After completing his education in the **United States**, **Jayaprakash Narayan** returned to India in **1929** with a strong intellectual influence of **Karl Marx**. During his journey back, he met several communist leaders in **London** and **India** and discussed issues related to India's freedom struggle and social revolution. However, he did not support the Indian communists' opposition to the **Indian National Congress**, which was leading the movement for India's independence.
- At the invitation of **Jawaharlal Nehru**, Jayaprakash Narayan joined the **Indian National Congress** in **1929** and soon became an active participant in the freedom movement. He was imprisoned in **1932** for his role in the **Civil Disobedience Movement**. During his imprisonment in **Nasik Jail**, he came into close contact with prominent socialist leaders such as **Ram Manohar Lohia**, **Ashok Mehta**, **Minoo Masani**, **Achyut Patwardhan**, and **C. K. Narayanswami**.
- These interactions strengthened his socialist ideology and led him to join the **Congress Socialist Party (CSP)**, headed by **Acharya Narendra Dev**, which functioned as the socialist wing within the Indian National Congress. As the **General Secretary of the CSP**, Jayaprakash Narayan, in **December 1939**, urged Indians to use the opportunities created by the **Second World War** to intensify the struggle against British rule and end colonial exploitation.

Q 109. Ans. C

Explanation:

- **Shri Krishna Singh**, popularly known as **Shri Babu and Bihar Kesari**, was the **first Chief Minister of Bihar** and is regarded as one of the principal architects of modern Bihar, alongside **Rajendra Prasad** and **Anugrah Narayan Sinha**. A prominent freedom fighter, he earned the title "**Bihar Kesari**" for his powerful speeches and leadership during

the national movement. He was first imprisoned in **1922** and spent nearly **eight years** in jail over the course of the freedom struggle.

- As Bihar's first Chief Minister, **Shri Krishna Singh** played a pivotal role in implementing land reforms by **abolishing the Zamindari system**. He also championed social equality by leading the **entry of Dalits into the Baidyanath Dham Temple at Baidyanath Temple**. He passed away on **31 January 1961**. In recognition of his outstanding contributions, the **India Post** issued a **commemorative postage stamp of ₹5 (500 paise)** in his honour in **2016**.

Q 110. Ans. C

Explanation:

- **Swami Sahajanand Saraswati**, the pioneering ascetic and architect of India's peasant movement, is best known for his fierce battle-cries and philosophical slogans that galvanized millions of farmers against British imperialists and oppressive zamindars.
 - ✓ "**Kaise Logey Malguzari, Latth Hamara Zindabad**" (How will you collect rent when our sticks are strong?) — Symbol of resistance against zamindari oppression.
 - ✓ "**Danda Mera Zindabad**" (Long live my staff) — The bamboo staff represented peasant unity and resistance.
 - ✓ "**Jo ann, vastr upjaega, ab so kanoon banaega**" (Those who produce food and clothing will now make the laws) — Demand for political power for peasants.
 - ✓ "**Yeh Bharatvarsh usi ka hai, ab shasan wahi chalayega**" (This India belongs to them; now they will govern it) — Assertion of peasant sovereignty and self-rule.

Q 111. Ans. D

Explanation:

- **Statement 1 is correct:** The Reserve Bank of India (RBI) was established on **1 April 1935** under the Reserve Bank of India Act, 1934. Initially, it was headquartered in Kolkata and was later permanently shifted to Mumbai in 1937. The RBI was originally constituted as a shareholders' bank.
- **Statement 2 is correct:** The RBI was nationalised on **1 January 1949** through the Reserve Bank (Transfer to Public Ownership) Act, 1948. After nationalisation, ownership passed entirely to the Government of India. The RBI serves as the banker to the government and is the sole authority for the issuance of banknotes in India (except the

one-rupee note issued by the Government of India).

- **Statement 3 is correct:** The Repo Rate is one of the RBI's most important monetary policy instruments. It is the rate at which commercial banks borrow short-term funds from the RBI against government securities. Since 2016, the repo rate has been decided by the Monetary Policy Committee (MPC), which aims to maintain inflation at 4% (+/- 2%) under the flexible inflation-targeting framework.

Q 112. Ans. D

Explanation:

- According to the World Trade Organization World Trade Outlook and Statistics 2025, the United States was the largest merchandise importer in the world, accounting for approximately 13.6% of global merchandise imports.
- **In contrast to imports:**
 - ✓ China remained the largest merchandise exporter in the world.
 - ✓ The United States continued to be the world's largest market for imported goods.

Q 113. Ans. B

Explanation:

- The National Statistical Office (NSO) is responsible for preparing and publishing the official estimates of national income, Gross Domestic Product (GDP), Gross Value Added (GVA), and other macroeconomic aggregates in India.
- **The NSO functions under the Ministry of Statistics and Programme Implementation (MoSPI) and releases:**
 - ✓ Annual National Income Estimates
 - ✓ Quarterly GDP Estimates
 - ✓ Consumer Expenditure and other statistical reports

Q 114. Ans. A

Explanation:

- A Giffen Good is a special type of inferior good for which demand increases when its price increases and decreases when its price decreases, thereby violating the normal Law of Demand.
- This occurs because the negative income effect of a price rise is so strong that it outweighs the substitution effect.

Q 115. Ans. D

Explanation:

- **Statement 1 is correct:** The Planning Commission

was replaced by NITI Aayog on 1 January 2015. NITI Aayog was established to promote cooperative federalism and serve as a policy think tank for the Government of India.

- **Statement 2 is correct:** The First Five Year Plan (1951–56) was based on the Harrod-Domar Growth Model, which emphasized increasing investment and capital formation to accelerate economic growth. The plan gave top priority to agriculture, irrigation, and energy in view of food shortages and the challenges faced after Partition.
- **Statement 3 is correct:** The Second Five Year Plan (1956–61) was based on the Mahalanobis Model and focused on the rapid development of heavy and basic industries. It emphasized the expansion of the public sector and the creation of capital goods industries to achieve long-term industrial growth.

Q 116. Ans. B

Explanation:

- Fiscal Deficit refers to the excess of the government's total expenditure over its total receipts excluding borrowings during a financial year.
 - ✓ $\text{Fiscal Deficit} = \text{Total Expenditure} - (\text{Revenue Receipts} + \text{Non-Debt Capital Receipts})$
- It represents the amount the government needs to borrow to meet its expenditure commitments.

Q 117. Ans. A

Explanation:

- **Statement 1 correct:** The Wholesale Price Index (WPI) measures changes in the prices of goods traded in bulk at the wholesale level, before they reach consumers. It is compiled and released by the Office of Economic Adviser, Department for Promotion of Industry and Internal Trade (DPIIT), Ministry of Commerce & Industry.
 - ✓ **Important Fact:**
 - ✓ **Base Year of WPI:** 2011–12
 - ✓ WPI covers Goods only and does not include services.
- **Statement 2 is correct:** The Consumer Price Index (CPI) measures changes in the prices paid by consumers at the retail level for a basket of goods and services.
- **Important Fact:**
 - ✓ CPI is compiled by the National Statistical Office (NSO).
 - ✓ **Base Year of CPI (Combined):** 2012
 - ✓ CPI includes both goods and services such as food, housing, education, health, and transport.

- **Statement 3 incorrect:** Before 2014, the RBI primarily looked at WPI. However, following the recommendations of the Urjit Patel Committee, the RBI shifted its policy target to CPI (Combined) in 2014. The committee argued that WPI reflects only the producer/wholesale price and ignores the cost of services (which make up a huge portion of the Indian economy). CPI, being retail-based, is a much better indicator of the inflation actually felt by the common citizen (the “cost of living”).
- **Important Facts:**
 - ✓ **Inflation Target:** 4%
 - ✓ **Tolerance Band:** $\pm 2\%$ (i.e., 2% to 6%)
 - ✓ The target is monitored by the Monetary Policy Committee (MPC).

Q 118. Ans. D

Explanation:

- Suresh Tendulkar Committee and the C. Rangarajan Committee were constituted to review and improve the methodology for estimating poverty in India.
 - ✓ **Tendulkar Committee (2005)**
 - ✓ The Tendulkar Committee was appointed by the Planning Commission to recommend a new poverty estimation methodology.
 - ✓ **Important Recommendations:**
 - Shifted away from the earlier calorie-based poverty line.
 - Considered expenditure on health, education, clothing, and shelter.
 - Used Mixed Reference Period (MRP) data from consumption surveys.
 - Estimated India’s poverty ratio at 21.9% in 2011–12.
- Appointed in 2012 (reporting in 2014), Rangarajan committee was a response to the controversy over the Tendulkar Committee’s “too low” poverty lines. Rangarajan maintained the consumption basket approach but significantly raised the poverty line, arguing that the Tendulkar estimates were insufficient for urban and rural basic needs.

Q 119. Ans. C

Explanation:

- The Charter Act of 1853 introduced, for the first time, an open competition system for the recruitment and selection of civil servants, throwing the covenanted civil service open to Indians as well, and the Macaulay Committee was appointed in 1854 to give effect to this principle.
- The Act separated, for the first time, the legislative

and executive functions of the Governor General’s council by adding six new members, called legislative councillors, to it, thereby creating a separate Central Legislative Council that came to function as a mini Parliament.

- The Charter Act of 1853 was the last of the series of Charter Acts and, unlike the previous ones, it did not specify any particular period for the continuance of the Company’s rule, an indication that the Company’s rule could be terminated at any time. It also introduced for the first time a local representation in the Central Legislative Council, paving the way for the later constitutional developments under the Crown.

Q 120. Ans. A

Explanation:

- Justice – Social, economic and political.
 - ✓ The Preamble secures justice in the social, economic, and political spheres for all citizens, an ideal drawn from the Russian Revolution and meaning the absence of privileges and the upliftment of the weaker sections.
- Liberty – of thought, expression, belief, faith and worship.
 - ✓ The Preamble secures liberty of thought, expression, belief, faith, and worship, an ideal drawn from the French Revolution, meaning the absence of restraints on individual activity within the bounds of law.
- Equality – of status and of opportunity.
 - ✓ The Preamble secures equality of status and of opportunity, which involves the absence of special privileges for any section and the provision of adequate opportunities for all without discrimination.
- Fraternity – assuring the dignity of the individual and the unity and integrity of the Nation
 - ✓ Fraternity means a sense of brotherhood, and the Preamble seeks to promote it by assuring the dignity of the individual and the unity and integrity of the nation, the word integrity having been added by the 42nd Amendment. The ideals of liberty, equality, and fraternity in the Preamble were taken from the French Revolution.

Q 121. Ans. C

Explanation:

- Under Article 33, Parliament can restrict or abrogate the fundamental rights of the members of the armed

forces, forces charged with the maintenance of public order (police forces), intelligence agencies, and analogous services, so as to ensure the proper discharge of their duties and the maintenance of discipline among them.

- Article 34 does not authorise the proclamation of martial law; the Constitution nowhere expressly provides for or defines martial law. Article 34 merely provides that when martial law is in force in any area, Parliament may by law indemnify government servants for acts done in connection with the maintenance or restoration of order, and may validate sentences passed and other acts done under martial law.
- The concept of martial law is implicit in the Constitution and is borrowed from English common law, where it means the suspension of ordinary law and the temporary government of a disturbed area by military tribunals. It differs from a national emergency, since martial law affects only fundamental rights in a localised area, suspends the operation of ordinary courts, and is imposed to restore order in a breakdown of law and order.

Q 122. Ans. B

Explanation:

- **Statement 1 is incorrect:** The Directive Principles do not have legal sanction behind them; their backing is moral and political, since they are non justiciable and cannot be enforced by the courts, though Article 37 declares them fundamental in the governance of the country and a duty of the State to apply them in making laws.
- **Statement 2 is correct:** Many directive principles have been implemented through legislation and programmes, such as land reform laws, the laws on minimum wages and equal remuneration, the Panchayati Raj system, the legal services authorities for free legal aid, environment protection laws, and employment guarantee schemes, which give practical effect to the goals set out in Part IV.
- The Directive Principles aim at the establishment of a welfare state and a social and economic democracy, as distinct from the political democracy guaranteed by the fundamental rights. They have been criticised for lacking legal force, for being illogically arranged without a consistent philosophy, and for being constitutionally superfluous, yet they serve as a moral and political yardstick for the conduct of the government and as a beacon for the courts in interpreting the laws.

Q 123. Ans. A

Explanation:

- **Statement 1 is correct:** Public interest litigation relaxes the traditional requirement of locus standi, under which only a person whose own right was infringed could approach the court, by permitting any public spirited person to seek relief on behalf of those who cannot approach the court themselves.
- **Statement 2 is correct:** Under the epistolary jurisdiction developed by the courts, a letter or even a postcard addressed to the court, or a news report highlighting a violation of rights, can be treated as a writ petition, so that procedural technicalities do not defeat the cause of justice for the disadvantaged.
- **Statement 3 is incorrect:** Public interest litigation cannot be used to settle a private dispute between two parties, since it is intended to vindicate a public cause involving the rights of the poor, the oppressed, or the community at large; the courts have framed guidelines to prevent its misuse as private interest, publicity interest, or political interest litigation. The Hussainara Khatoon case is regarded as the first reported public interest litigation in India.

Q 124. Ans. A

Explanation:

- The Constitution does not prescribe any specific procedure for the selection and appointment of the Chief Minister; Article 164 merely provides that the Chief Minister shall be appointed by the Governor, and by convention the Governor appoints the leader of the majority party in the legislative assembly.
- A person who is not a member of the state legislature can be appointed Chief Minister, provided that he secures membership of either House of the state legislature within six months of his appointment, failing which he ceases to be the Chief Minister.
- Where no party has a clear majority, the Governor exercises personal discretion in appointing the Chief Minister, usually selecting the person most likely to command the confidence of the assembly. The Chief Minister can be a member of either House of the state legislature where the state has two Houses, and the Council of Ministers is collectively responsible to the legislative assembly.

Q 125. Ans. D

Explanation:

- Article 348 provides that, until Parliament otherwise provides, the proceedings in the Supreme Court

and in every High Court shall be in the English language, as shall the authoritative texts of all bills, acts, ordinances, orders, rules, regulations, and byelaws.

- Article 351 directs the Union to promote the spread of the Hindi language and to develop it so that it may serve as a medium of expression for all the elements of the composite culture of India, by drawing on Sanskrit and the other languages of the country.
- Article 343 declares Hindi in the Devanagari script to be the official language of the Union, while allowing the continued use of English; the Official Languages Act of 1963 permits the indefinite continuation of English. Article 344 provides for the appointment of an Official Language Commission and a Committee of Parliament on Official Language, and Article 350B provides for a Special Officer for Linguistic Minorities.

Q 126. Ans. C

Explanation:

- **Assertion is true:** The Election Commission conducts elections through a vast machinery that is mobilised at election time, headed in each state by a Chief Electoral Officer and extending down through District Election Officers, Returning Officers, and Presiding Officers at the polling stations.
- **Reason is false:** The officials of this election machinery are not permanent employees of the Election Commission; they are officers of the central and state governments who are deputed for election work and who function under the superintendence, direction, and control of the Commission during that period. The Election Commission itself has only a small permanent secretariat at the central level.
- The Chief Electoral Officer of a state is nominated by the Election Commission in consultation with the state government and supervises the election work in the state. The District Election Officer supervises the work in the district, while the Returning Officer is responsible for the conduct of the election in a particular constituency, all of them being drawn from the existing administrative apparatus and placed at the disposal of the Commission.

Q 127. Ans. A

Explanation:

- Under Article 75 the Prime Minister is appointed by the President, who by convention appoints the leader of the majority party in the Lok Sabha; where no

party has a clear majority, the President may exercise personal discretion in appointing the person most likely to command the confidence of the House.

- The Prime Minister need not necessarily be a member of the Lok Sabha; a member of the Rajya Sabha can also be appointed Prime Minister, and several Prime Ministers have been members of the upper House. A person who is not a member of either House can be appointed Prime Minister, provided he secures membership of either House within six months.
- The Prime Minister is the head of the government, the leader of the Council of Ministers, and the principal channel of communication between the President and the Council. The Prime Minister recommends the persons who are appointed as ministers, allocates and reshuffles their portfolios, and can recommend the dissolution of the Lok Sabha to the President, occupying a position of exceptional authority in the parliamentary system.

Q 128. Ans. A

Explanation:

- **Statement 1 is correct:** Under Article 165 the Advocate General is the highest law officer of the state, corresponding to the Attorney General of India at the Centre, and is appointed by the Governor from among persons qualified to be appointed a judge of a High Court, to advise the state government on legal matters and to perform such legal duties as may be assigned.
- **Statement 2 is incorrect:** The Advocate General does not hold office for a fixed term of five years; the Constitution does not fix the term of office, and the Advocate General holds office during the pleasure of the Governor and may be removed by the Governor at any time, and conventionally resigns when the government that appointed him resigns.
- The Advocate General has the right to speak and to take part in the proceedings of both Houses of the state legislature and any committee of which he is named a member, but without the right to vote, and he enjoys all the privileges and immunities of a member of the state legislature. He is entitled to audience in all courts within the state in the performance of his official duties.

Q 129. Ans. C

Explanation:

- The Gulf Stream is a warm, swift, and powerful ocean current that flows along the eastern coast of

North America. Originating in the Gulf of Mexico, it passes through the Strait of Florida and moves northward along the eastern coast of the United States before crossing the North Atlantic Ocean. It is one of the most significant ocean currents in the world and plays a crucial role in regulating the climate of north-western Europe.

- As the Gulf Stream moves northeastward, it continues as the North Atlantic Drift, carrying warm waters towards the British Isles and Western Europe, thereby moderating winters in these regions despite their high latitudes.

Q 130. Ans. B

Explanation:

- **Chinook:** Correctly matched to the **Rocky Mountains, North America**. The Chinook is a warm, dry wind that blows down the eastern slopes of the Rockies. It is famously known by Native Americans as the “Snow Eater” because its sudden arrival can melt thick blankets of snow in a matter of hours due to adiabatic heating.
- **Mistral:** Correctly matched to **France**. The Mistral is a strong, cold, and dry north-westerly wind that accelerates as it funnels through the Rhone Valley in southern France, ultimately blowing out into the Mediterranean Sea. It is most frequent in winter and spring.
- **Loo:** Correctly matched to **Northern India**. The Loo is a notorious, extremely hot, and dry summer wind that blows across the Indo-Gangetic plains of Northern India and Pakistan during May and June. Exposure to it can cause severe heatstroke.
- **Foehn:** Correctly matched to **The Alps**. The Foehn (or Föhn) is a warm, dry, downslope wind that occurs on the leeward (downwind) side of the Alps in Europe. It shares the exact same meteorological mechanism as the Chinook, often bringing sudden warming to the valleys of Switzerland, Austria, and Germany.

Q 131. Ans. B

Explanation:

- A fault line is the line or plane along which displacement of fractured rock strata takes place during faulting. Faulting occurs when tectonic forces (tensional, compressional, or shear stresses) exceed the strength of rocks, causing them to fracture and move relative to one another. The movement may be vertical, horizontal, or oblique, producing characteristic landforms such as rift

valleys, block mountains, fault scarps, and linear valleys.

- A fault is a fracture in the Earth’s crust along which rocks have undergone relative displacement.
- **Major types of faults include:**
 - ✓ Normal Fault – caused by tensional forces; produces rift valleys (grabens).
 - ✓ Reverse (Thrust) Fault – caused by compressional forces; common in fold mountain regions.
 - ✓ Strike-slip (Transform) Fault – caused by shearing forces; movement is predominantly horizontal.

Q 132. Ans. C

Explanation:

- The “Furious Fifties” refers to the oceanic belt between 50°S and 60°S latitudes in the Southern Hemisphere. This region is renowned for its extremely strong and persistent westerly winds, which blow almost uninterrupted around the globe because there are very few landmasses to obstruct their flow.
- The absence of significant continental barriers allows these winds to attain very high speeds, generating powerful ocean currents, high waves, and frequent storms.
- **Additional Information:** The belts of strong westerly winds in the Southern Hemisphere are:
 - ✓ **Roaring Forties:** 40°S – 50°S
 - ✓ **Furious Fifties:** 50°S – 60°S
 - ✓ **Screaming Sixties:** 60°S – 70°S

Q 133. Ans. A

Explanation:

- The Bosphorus Strait, which connects the Black Sea to the Sea of Marmara, has long been known among sailors as the “Throat of Constantinople.” The name reflects its strategic position beside the historic city of Constantinople (present-day Istanbul) and its role as a narrow maritime gateway between the Black Sea and the Mediterranean Sea.
- The Bosphorus is one of the world’s most strategically important straits, separating Europe and Asia and serving as a vital international shipping route for countries bordering the Black Sea.

Q 134. Ans. B

Explanation:

- **Assertion (A) is true:** Materials in the Earth’s deeper interior, particularly in the asthenosphere

(upper mantle), are believed to exist in a plastic or semi-molten state. Although temperatures are extremely high, the immense pressure prevailing at great depths prevents complete melting. As a result, rocks behave plastically, allowing them to flow slowly over geological time rather than remaining completely liquid. This plastic behaviour facilitates mantle convection, which drives plate tectonics.

- **Reason (R) is true:** The temperature of the Earth increases with depth, a phenomenon known as the geothermal gradient. On average, the temperature rises by about 25–30°C per kilometre in the Earth's crust, although the rate varies with depth and geological conditions.
- **Why (R) is not the correct explanation of (A):**
 - ✓ While the increase in temperature is a true statement, it does not by itself explain why deep Earth materials remain in a plastic state. Their plastic behaviour results from the combined effect of very high temperature and enormous pressure. High temperatures tend to melt rocks, whereas the equally high pressures at depth inhibit complete melting, producing a plastic (ductile) state instead of a fully molten one

Q 135. Ans. A

Explanation:

- **Statement 1 is correct:** The Richter Scale measures the magnitude of an earthquake, which represents the amount of energy released at the earthquake's focus. Developed by Charles F. Richter in 1935, it is a logarithmic scale, meaning that each whole-number increase represents a tenfold increase in the amplitude of seismic waves and approximately 31.6 times more energy released than the previous magnitude.
- **Statement 2 is incorrect:** The Modified Mercalli Intensity (MMI) Scale does not rely on instrumental recordings of seismic waves. Instead, it measures the intensity of an earthquake based on its observed effects, such as damage to buildings, changes in the natural environment, and people's experiences. The scale ranges from I (Not Felt) to XII (Total Destruction). Instrumental recordings are used to determine magnitude, not Mercalli intensity.

Q 136. Ans. C

Explanation:

- Drumlins are elongated, smooth, oval-shaped glacial landforms composed primarily of glacial

till, an unsorted and unstratified mixture of clay, silt, sand, gravel, and boulders deposited directly by moving glacier ice. They are formed beneath continental glaciers or ice sheets through the processes of glacial deposition and reshaping.

- A typical drumlin has a steep, blunt end facing the direction from which the glacier advanced (stoss side) and a gentle, tapered end pointing in the direction of ice movement (lee side).

Q 137. Ans. A

Explanation:

- **Statement 1 is correct:** Cinder cone volcanoes are the simplest and smallest type of volcanic cone, formed by the accumulation of loose pyroclastic materials, mainly cinders (scoria), volcanic ash, and lapilli, around a single volcanic vent. These volcanoes are characterized by steep slopes (generally 30°–40°) and a bowl-shaped summit crater. They are usually produced during explosive eruptions of gas-rich basaltic or andesitic magma.
- **Statement 2 is incorrect:** Cinder cone volcanoes are extrusive volcanic landforms, not intrusive igneous rocks. They are formed by the deposition of volcanic fragments on the Earth's surface during eruptions. Intrusive igneous rocks, such as granite, diorite, and gabbro, crystallize beneath the Earth's surface from slowly cooling magma and do not form volcanic cones.

Q 138. Ans. D

Explanation:

- **Statement 1 is correct:** The Indus, the Sutlej, and the Brahmaputra all have their sources in the immediate vicinity of the Mount Kailash–Lake Manasarovar region in Tibet, within roughly 60–100 km of each other. The Indus originates from glaciers north of Kailash, the Sutlej rises near the Manasarovar–Rakshastal lakes to the west of Kailash, and the Brahmaputra (as the Yarlung Tsangpo) originates on the eastern slope of Kailash. All three are antecedent rivers, meaning they are older than the Himalayas and have cut deep transverse gorges as the mountains rose beneath them.
- **Statement 2 is correct:** The Tsangpo River, after flowing eastward across southern Tibet for nearly 1,800 km, takes a sharp southward turn around the Namcha Barwa massif, the eastern syntaxis of the Himalayas. In doing so, it carves out the Tsangpo Gorge (Yarlung Tsangpo Grand Canyon), widely regarded as the deepest canyon in the world. After

this dramatic bend, the river enters India as the Siang, later becoming the Dihang and eventually the Brahmaputra.

- **Statement 3 is correct:** Unlike the Indus, Sutlej, and Brahmaputra, which all originate north of the Great Himalayan Range in the Tibetan Plateau and cut through the range via antecedent drainage, the Ganga's principal source lies within the Great Himalayas themselves. Its main headstreams, the Bhagirathi (from the Gangotri glacier) and the Alaknanda, rise on the southern slopes of the Great Himalayan Range in Uttarakhand, well south of the Trans-Himalayan zone where the other three rivers begin. This makes the Ganga the only major Himalayan river system among the three whose principal source is not located in the Kailash-Manasarovar/Trans-Himalayan belt.

Q 139. Ans. A

Explanation:

- **Statement 1 is correct.** The North-Eastern Plateau Belt, covering Jharkhand, Odisha, and West Bengal, is the richest of all Mineral Belts of India, containing 100% of kyanite reserves, with significant deposits documented specifically at Lapsa Buru in the western part of the Singhbhum shear zone.
- **Statement 2 is correct.** Odisha accounts for about 98% of the total proved chromite reserves of the country, of which about 97% occur in the Sukinda Valley, and these deposits are predominantly stratiform in nature, with the chromiferous ultramafic rocks of Sukinda Valley intrusive into the Iron-Ore series, confirming the association with ultramafic rocks.
- **Statement 3 is incorrect.** Salem's magnesite deposits are genetically associated with the **Salem Ultramafic Complex / Chalk Hills ultramafic rocks**, formed through serpentinisation and carbonation of olivine-rich ultramafic rocks, not the Dharwar Supergroup. An association of olivine-serpentine-magnesite is widely distributed in the Salem Ultramafic Complex (SUC) in Southern India, with magnesite deposits of the Salem mafic-ultramafic Complex (SMUC) in southern India formed through alteration of ultramafic rocks by low-temperature fluids. The Dharwar Supergroup is a distinct geological classification (associated more with gold, iron ore in Karnataka's Dharwar craton) and is not the genetic source attributed to Salem's magnesite in the geological literature.

Q 140. Ans. A

Explanation:

- **Statement 1 is correct.** Weakening of the equatorial easterly trade winds is the primary trigger for El Niño, allowing the warm surface water pool, normally held in the western Pacific, to shift eastward across the equatorial Pacific.
- **Statement 2 is correct.** During El Niño, the thermocline deepens in the eastern equatorial Pacific as warm water accumulates there, suppressing the cold-water upwelling that normally occurs off the Peru and Ecuador coasts.
- **Statement 3 is incorrect.** This is the reverse of what actually happens. During El Niño, the Walker Circulation **weakens** (not strengthens), and convection **shifts away from** the western Pacific toward the central and eastern Pacific, following the relocated warm water. Enhanced convection over the western Pacific is characteristic of normal conditions or La Niña, not El Niño.

Q 141. Ans. C

Explanation:

- 'Ellora caves' is located in 'Maharashtra' in the same way 'Hampi' is located in 'Karnataka'.

Q 142. Ans. B

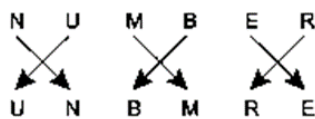
Explanation:

- **Formula to find the time of mirror image when:** time is given between 1 to 11 o'clock, time is subtracted by 11:60
- $11:60 - 05:27 = 06:33$

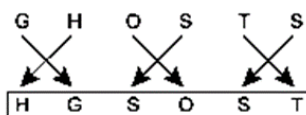
Q 143. Ans. C

Explanation:

- Since,



In the same way,



Q 144. Ans. C**Explanation:**

- The number of shops that sell only WATCHES and SHOES is 123.

Q 145. Ans. A**Explanation:**

- When, we write down the numbers of dots in clock wise direction starting from '1'
- In 1st figure, 1 5 3
- In 2nd figure, 1 2 4
- So, when the 3 dots face is at the top, then the 4 dots face is at bottom.

Q 146. Ans. C**Explanation:**

Given statements:

- $N \geq P > R = O < M \leq Q; K \geq L = N < S$
- On combining, we get
- $K \geq L = N \geq P > R = O < M \leq Q; S > N \geq P > R = O < M \leq Q$

Conclusions:

- **K > P:** False (As, $K \geq L = N \geq P$, so $K \geq P$)
- **$Q \geq S$:** False (As, $S > N \geq P > R = O < M \leq Q$, so relation between Q and S can't be determined)
- **$S > Q$:** False (As, $S > N \geq P > R = O < M \leq Q$, so relation between S and Q can't be determined)
- Conclusion II and III are complementary.
- Either conclusion II or III is true and I is not true.

Q 147. Ans. D**Explanation:**

$$\begin{aligned} &\Rightarrow \left(5\frac{3}{5} \div 2\frac{1}{10}\right) \div 3\frac{1}{5} \\ &\Rightarrow \left(\frac{28}{5} \div \frac{21}{10}\right) \div \frac{16}{5} \\ &\Rightarrow \left(\frac{28}{5} \times \frac{10}{21}\right) \times \frac{5}{16} \\ &\Rightarrow \frac{8}{3} \times \frac{5}{16} \\ &\Rightarrow \frac{5}{6} \end{aligned}$$

Q 148. Ans. C**Explanation:**

- The first five odd prime numbers are:
- 3, 5, 7, 11, 13
- Their sum = $3 + 5 + 7 + 11 + 13 = 39$
- Average = $39/5 = 7.8$
- Therefore, the required average = 7.8

Q 149. Ans. C**Explanation:**

Total food = $10 \times 40 = 400$ person-days

- Let the number of guests be = x
- Now, the total number of people = $(10 + x)$
- **According to the problem:**
- $(10 + x) \times 25 = 400$
- $\Rightarrow 10 + x = 16$
- $\Rightarrow x = 6$
- Therefore, 6 guests joined the family.

Q 150. Ans. B**Explanation:**

- Let the seven consecutive odd numbers be = n, (n+2), (n+4), (n+6), (n+8), (n+10), (n+12)
- The sum of the seven consecutive odd numbers = 315
- $\Rightarrow n + (n+2) + (n+4) + (n+6) + (n+8) + (n+10) + (n+12) = 315$
- $\Rightarrow 7n + 42 = 315$
- $\Rightarrow 7n = 273$
- $\Rightarrow n = 39$
- Third odd number = $39 + 4 = 43$
- Fifth odd number = $39 + 8 = 47$
- Required sum = $43 + 47 = 90$

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